



# Fiber Reinforced Shotcrete and Earth Retaining Structures for Rockfall Protection

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- ADOT-
  - Providing Safe, Quality Roadways for Arizona
  - Support the Economical Development of the region
  - Costs associated with road closure due to rockfall and ground instability are excessive, unpredictable and can have grave consequences for local industry, tourism, and commerce.
- ASU Research in construction Materials
  - Development of new technologies for direct implementation in infrastructure development

- Areas of applications for structural use of Fiber Reinforced concrete
  - Rock fall protection, and Slope Stability
  - Applications for Infrastructure development
- Development of Design procedures for Strain softening and hardening FRC
  - Material models, Moment-curvature relationship
  - load-deflection results compared with experiments
  - Toughness based Design
  - Use of post-peak residual strength
- Conclusions



## ■ News:

- Additional rock falls keep SR 264 closed
- Mudslide on state highway 89A South between Flagstaff and Sedona

## ■ Ongoing maintenance projects:

- Kingman District Construction Projects, November 2012
- I-40 MP 48, Rockfall Mitigation
- I-40 MP 102
- Cross Mountain—Jolly Road Rockfall Protection
- Rockfall prevention project set for State Route 78 in Greenlee County's Mule Creek Pass



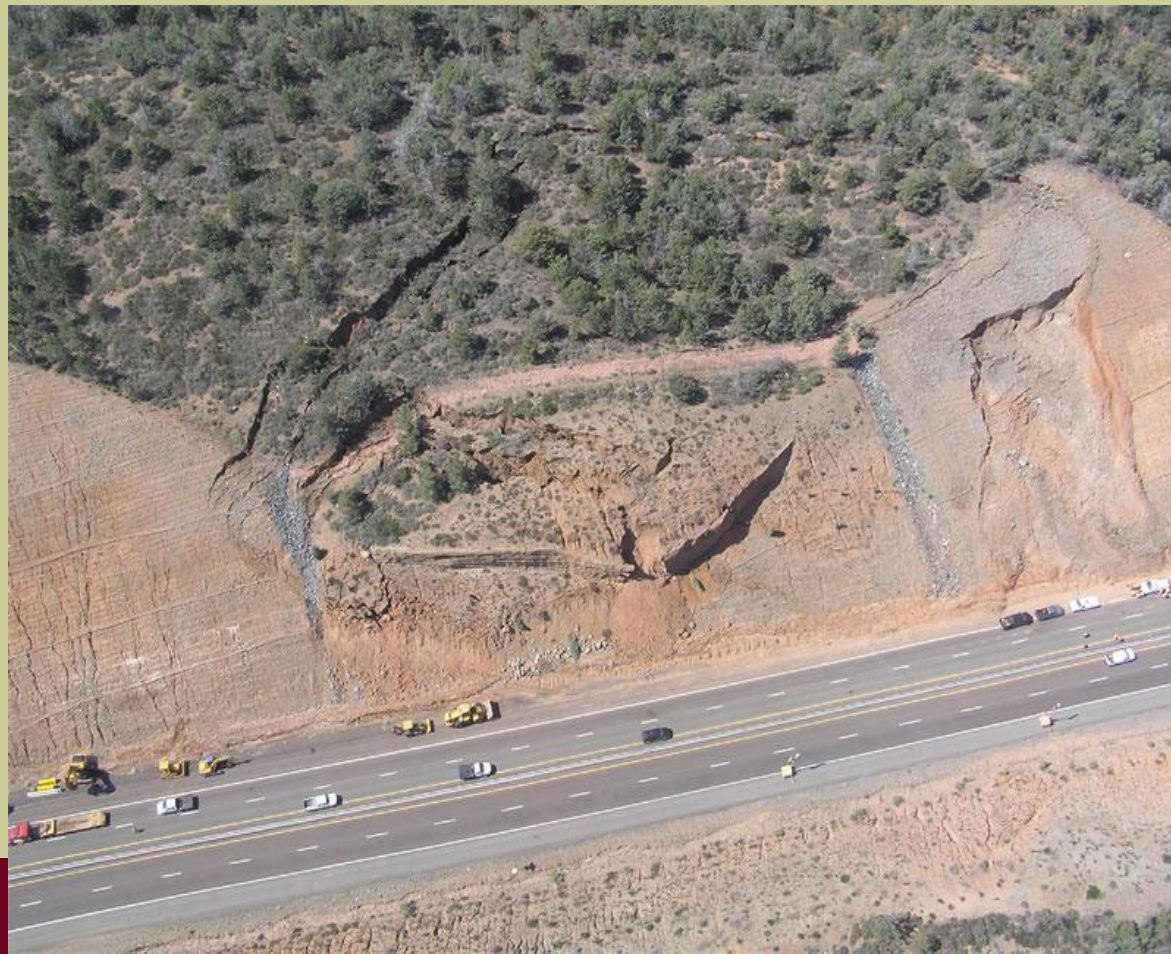
- “A large rock fall has closed State Route 264 southeast of Tuba City. State highway officials now say the earliest the highway will reopen is likely Thursday because the equipment necessary to remove the debris has to be driven up to northern Arizona” January 05, 2010
- “Arizona Department of Transportation crews started clearing the debris Monday but more rocks started falling, and the supervisor on the scene pulled his teams from the area for safety reasons. Then, another large section of the hill fell, covering more of the road.”





# STATE ROUTE 87 LANDSLIDES, Payson, Arizona

- Mitigation plans have been in effect during 2004, 2005, 2008, and 2009





# ASU- Rio Tinto Project – Magma Copper mine, Superior , Arizona

ASU has helped in evaluating the initial fiber-reinforced shotcrete support design for a 2000 m-deep shaft at Resolution Mine, Arizona.





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# Project Overview

- Evaluate fiber-reinforced shotcrete performance
- No rockbolts planned for initial support.
- Deep shaft (2189 m)
- Three geological units
- A range of stress conditions (increased with depth)
- Several modes of instability: gravity driven, rockmass shear yielding, brittle failure



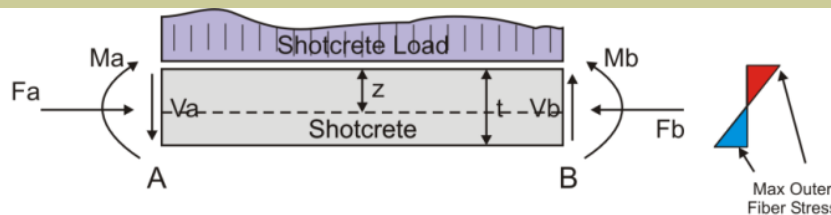
- *Shotcrete is applied prior to form setting for permanent lining*
- *No rock bolts planned for initial support*
- *Several modes of instability: gravity driven, rock mass, shear yielding, brittle failure*

*Tensile extreme fiber stress exceeding tensile strength of shotcrete*



*3DEC-predicted liner stresses and potential cracking in TAL unit*

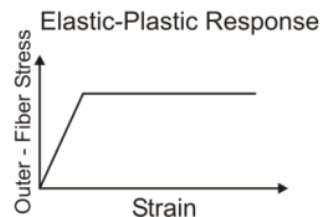
- Linear-elastic shotcrete liner installed in low stress, blocky ground.
- Early age properties use for shotcrete and rock-shotcrete interface along full length of model shaft.
- Extreme fiber stress used to evaluate potential shotcrete cracking.
- Tensile extreme fiber stress exceeding the tensile strength *combined* with tensile thrust at a given liner section indicates potential for a crack to form through the liner.
- Large regions of connected cracks would indicate potential for a block breaking through the shotcrete.



Combines Flexural Strength:

$$F/A \pm Mz/I \leq \sigma_{\text{ten, comp strength}}$$

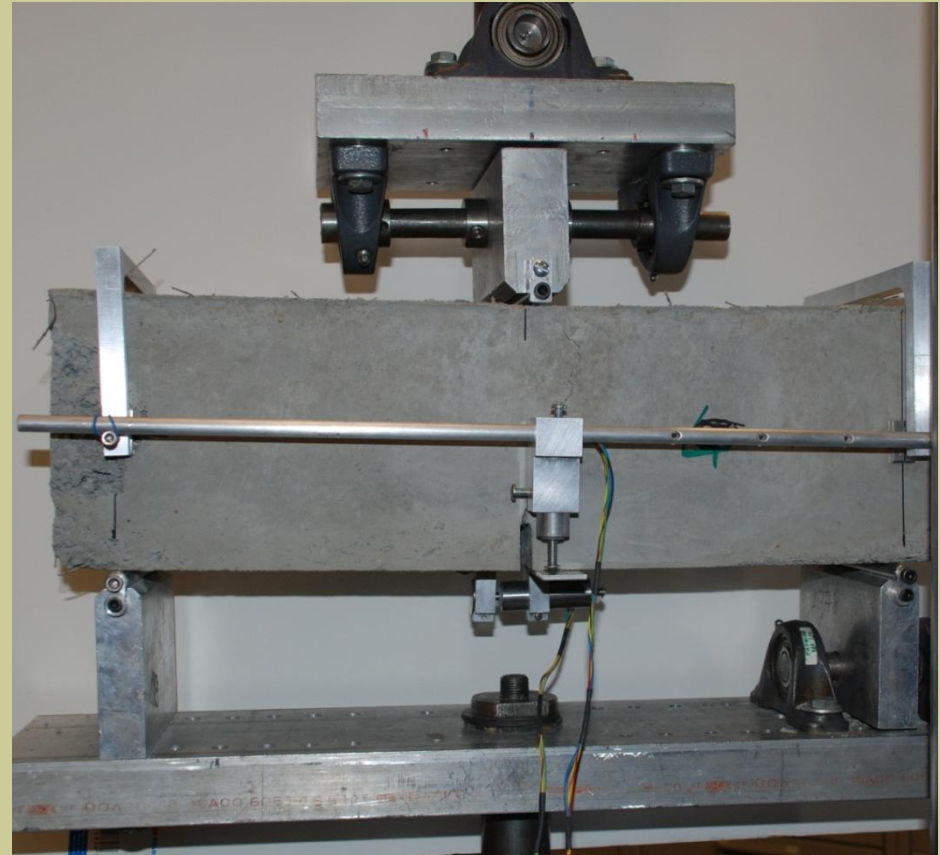
$$A = 1 \cdot t; I = 1 \cdot t^3/12$$

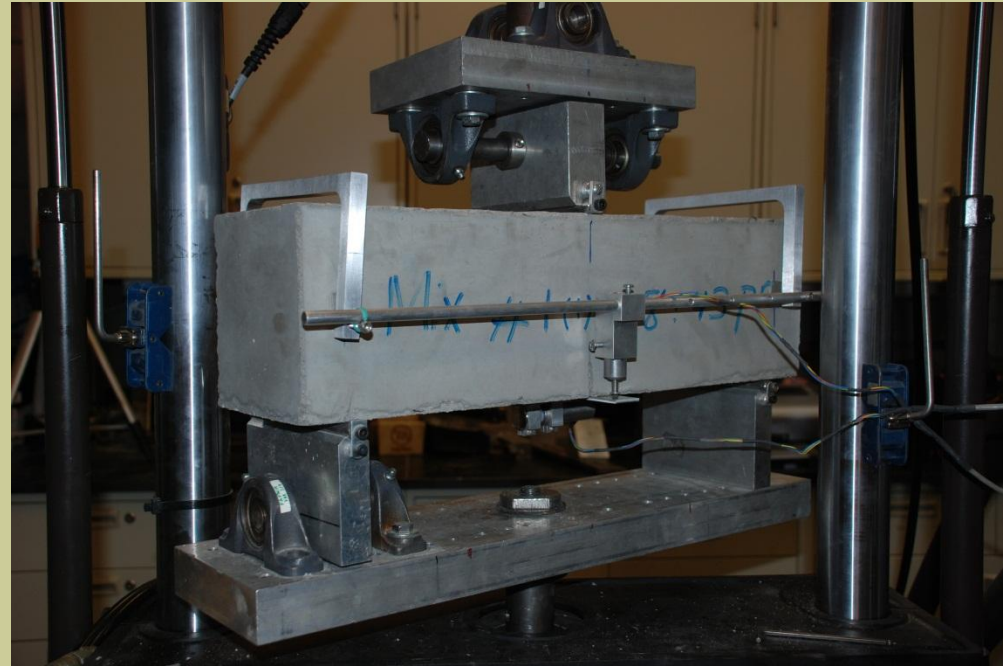
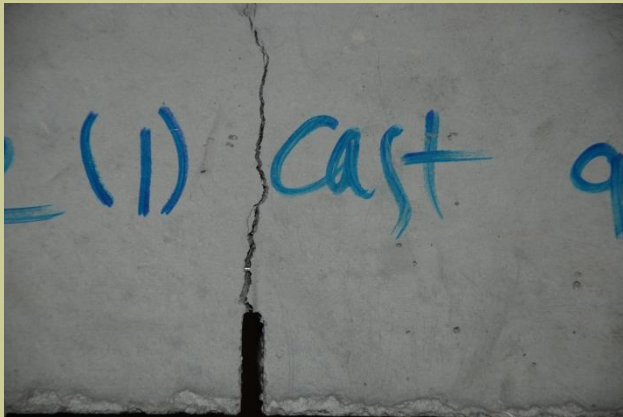
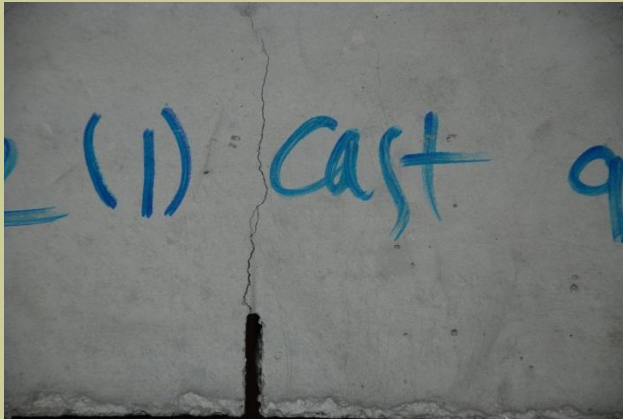


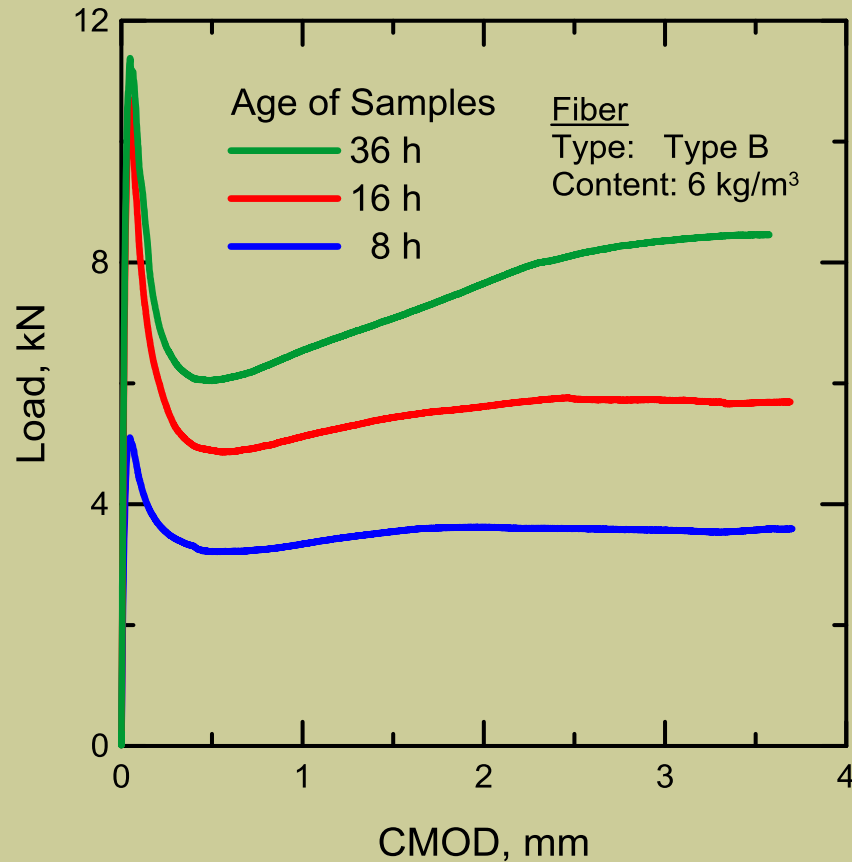
**Extreme Fiber Stress**



- Beam  $53 \times 15 \times 15 \text{ cm}^3$ , Span: 45 cm
- Initial notch length: 2.5 cm
- Load is applied along the notch at a single point
- Test performed under closed loop CMOD control at 0.12mm/min
- Results are correlate with Round Panel Tests for material modeling





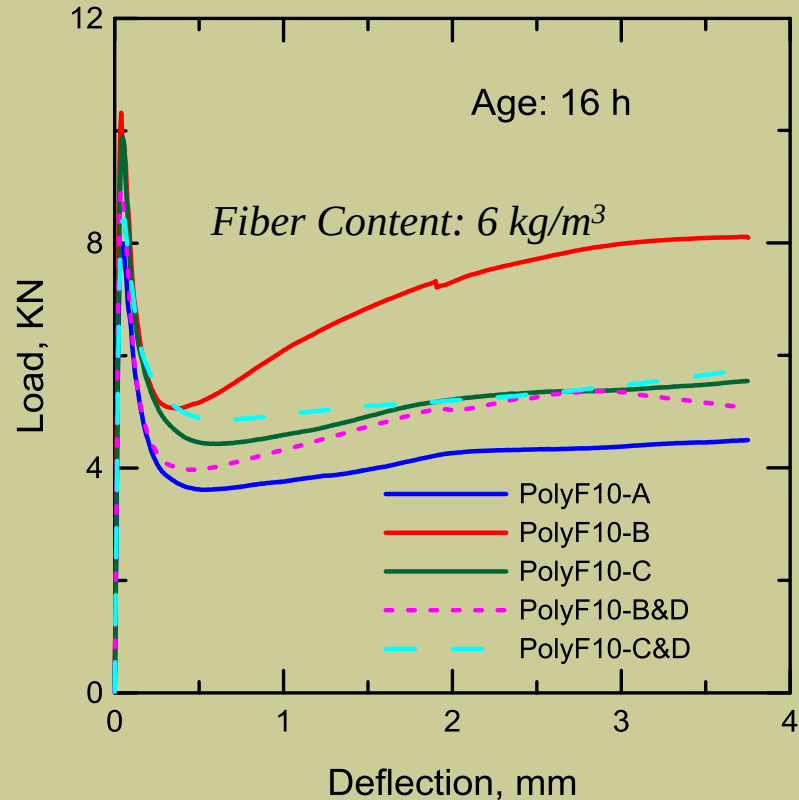


Mix Proportion (kg/m<sup>3</sup>)

| Mix ID           | PolyF10-TypeB     |
|------------------|-------------------|
| Portland Cement  | 475               |
| Fly Ash          | 60                |
| Silica Fume      | 15                |
| Fine Aggregate   | 1050              |
| Coarse Aggregate | 400               |
| Water            | 220               |
| Fiber            | 6                 |
| Fiber Type       | Type 3            |
| Accelerator (%)  | 4                 |
| w/cm             | 0.40              |
| s/cm             | 1.9               |
| F'c (28 d), Mpa  | 27.1 <sup>a</sup> |

| Age, h | K, KN/mm | CMOD at Max Load, mm | f <sub>b</sub> , MPa | Max CMOD, mm | Flex. Tough., KN.mm |
|--------|----------|----------------------|----------------------|--------------|---------------------|
| 8      | 176      | 0.0431               | 1.57                 | 3.8          | 8.05                |
| 16     | 273      | 0.0406               | 2.84                 | 3.8          | 12.28               |
| 36     | 257      | 0.0431               | 3.09                 | 3.8          | 14.60               |

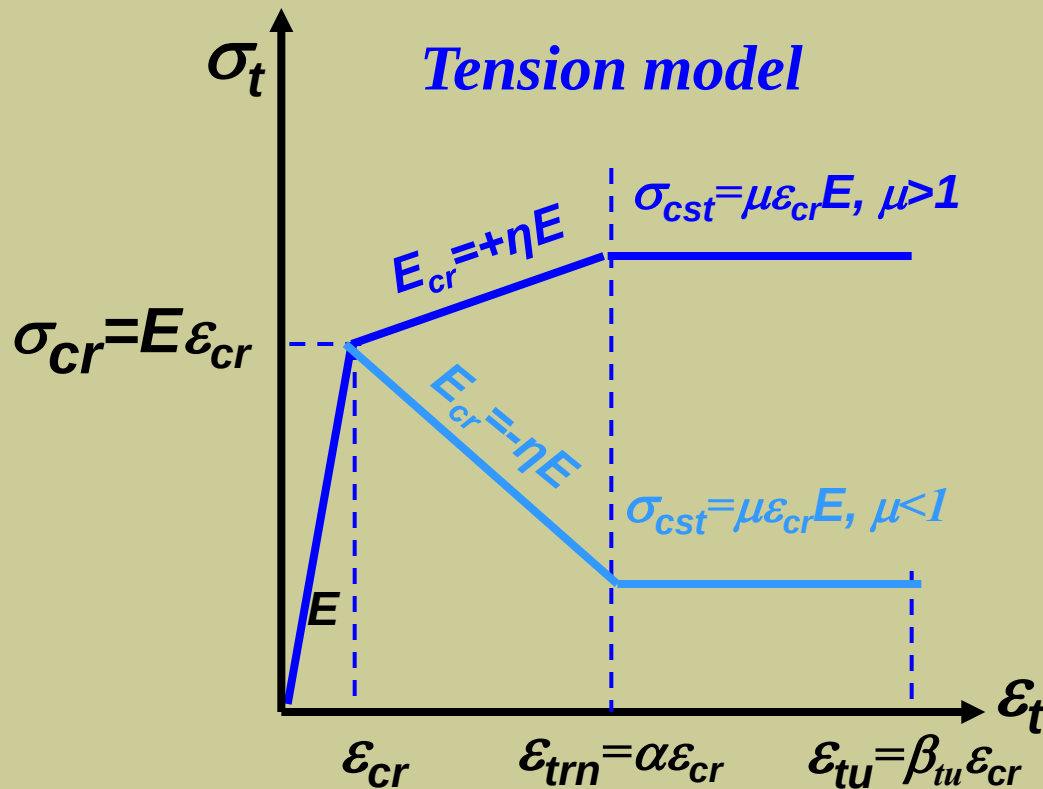
# Effect of Fiber Types (Polymeric)



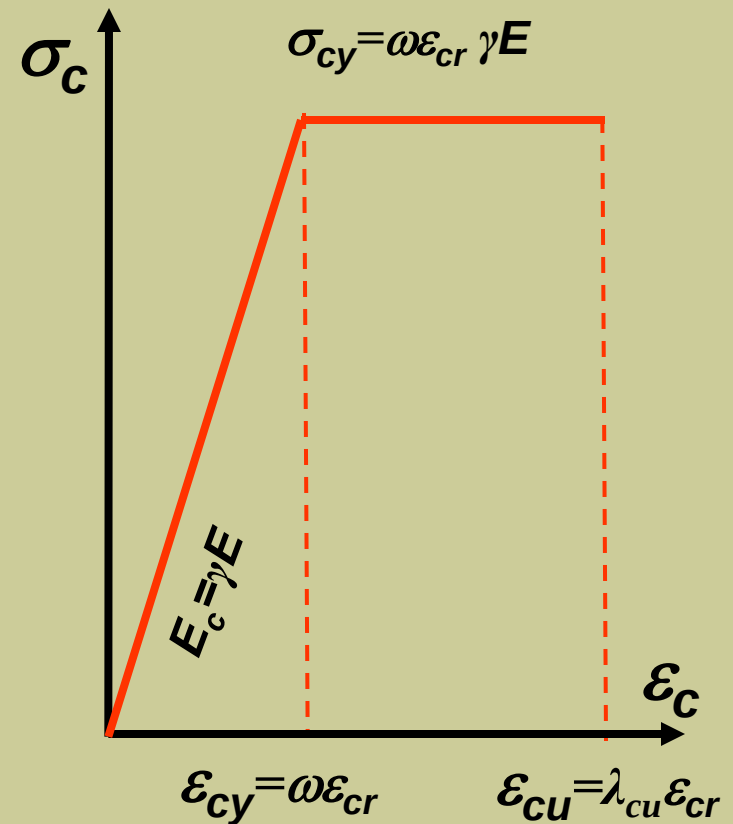
| Fiber Type  | Type A                                         | Type B          | Type C                       | Type D                          |
|-------------|------------------------------------------------|-----------------|------------------------------|---------------------------------|
| Base        | Monofilament Polypropylene/ Polyethylene blend | Modified Olefin | Modified Polypropylene Blend | fibrillated polypropylene fiber |
| S. Gravity  | 0.92                                           | 0.9-0.92        | 0.91                         | 0.91                            |
| E (GPa)     | 5                                              | 10              | 6.55                         | 5.5                             |
| UTS (MPa)   | 600 -650                                       | 550             | 655                          | 410                             |
| Length (mm) | 50                                             | 50              | 37.5                         | 50                              |

| Mix ID      | K, KN/mm | CMOD at Max Load, mm | f <sub>b</sub> , MPa | Max CMOD, mm | Flex. Tough., KN.mm | F <sub>c</sub> , MPa |
|-------------|----------|----------------------|----------------------|--------------|---------------------|----------------------|
| PolyF10-A   | 239      | 0.0457               | 2.43                 | 3.83         | 9.6                 | 29.0 <sup>a</sup>    |
| PolyF10-B   | 281      | 0.0431               | 2.76                 | 3.83         | 15.8                | 27.1 <sup>a</sup>    |
| PolyF10-C   | 251      | 0.0457               | 2.79                 | 3.82         | 12.3                | 28.8 <sup>a</sup>    |
| PolyF10-B&D | 269      | 0.0609               | 2.48                 | 3.82         | 13.8                | 28.7 <sup>a</sup>    |
| PolyF10-C&D | 266      | 0.0508               | 2.28                 | 3.81         | 14.0                | 25.8 <sup>a</sup>    |

- Material parameters are described as a multiple of the first cracking tensile strain ( $\varepsilon_{cr}$ ) and tensile modulus ( $E$ )



## Compression model

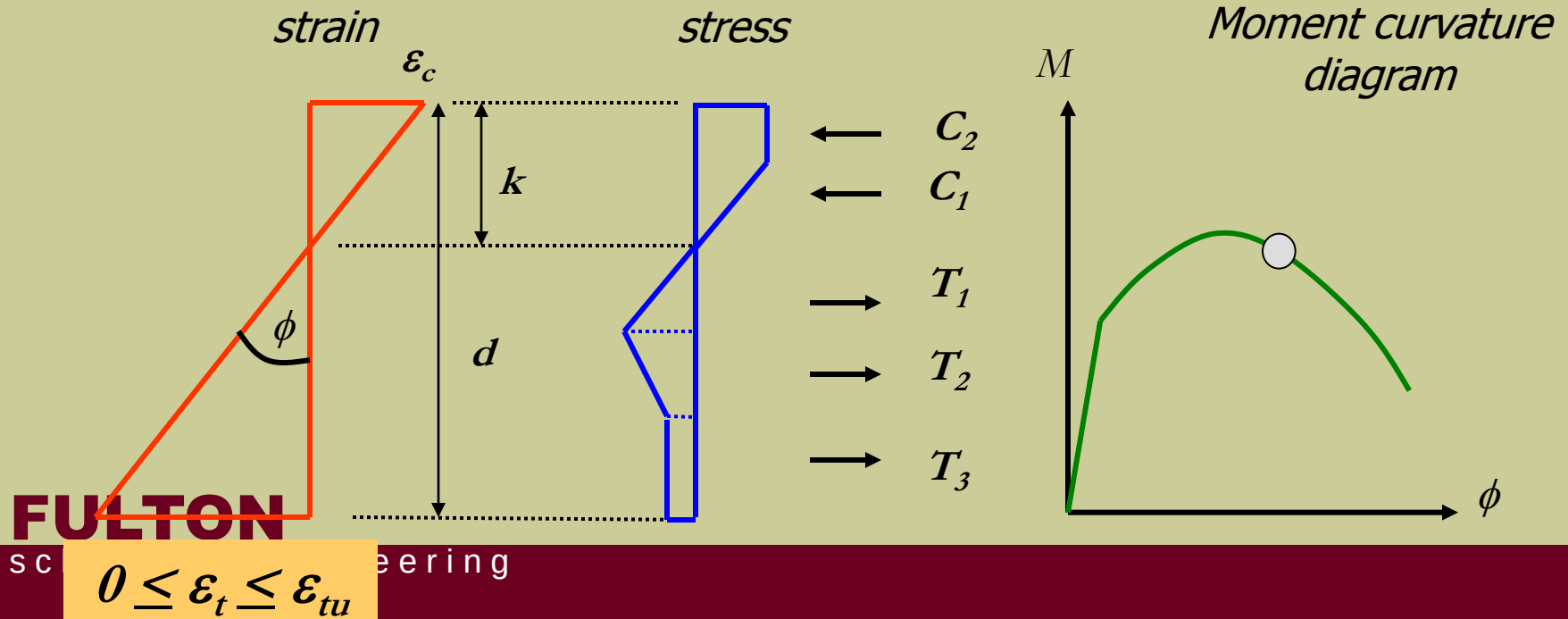


# Moment-Curvature Diagram

- Incrementally impose  $0 \leq \varepsilon_t \leq \varepsilon_{tu}$ 
  - Strain Distribution
  - Stress Distribution
- $\Sigma F = 0$ , determine  $k$  (Neutral axis)
  - $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$  and  $\phi = \varepsilon_c / kd$
- Normalization  $M' = M/M_0$  and  $\phi' = \phi / \phi_{cr}$

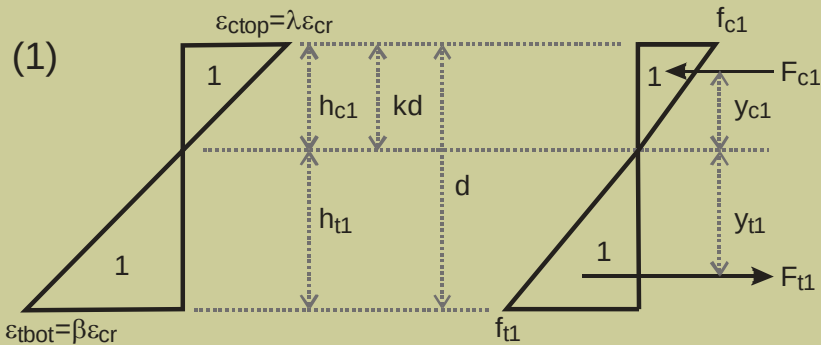
$$F_{c1} = b \int_0^{kd} f_{c1}(y) dy$$

$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} f_{c1}(y) y dy$$

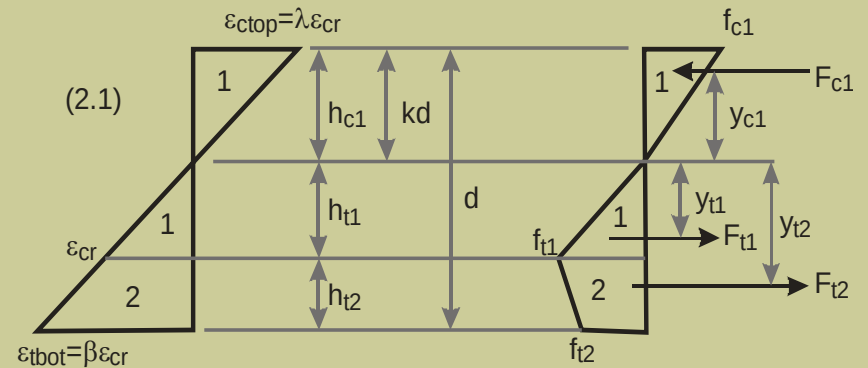


# Stress and Strain Distribution Tensile Regions 1.0, 2.1 and 3.1

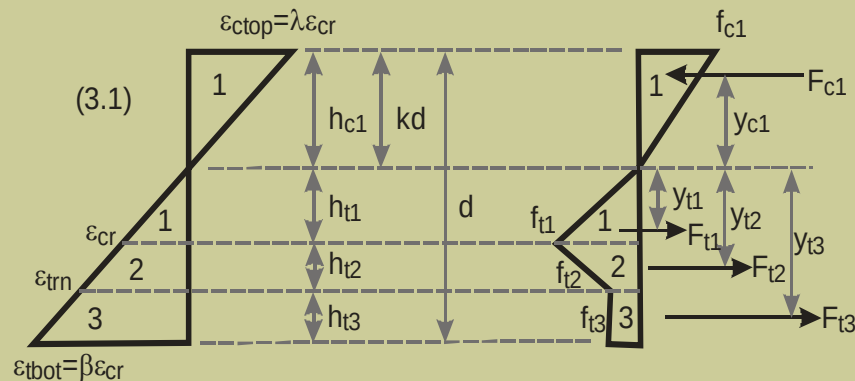
$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$



$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$



# Closed Form Solutions for Strain Hardening/Softening material

| Stage | k                                                                                                                                                                     | M'                                                                                                                                                                                                                       | $\phi'$                                  |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1     | $k_1 = \begin{cases} \frac{1}{2} & \text{for } \gamma=1 \\ \frac{-1+\sqrt{\gamma}}{-1+\gamma} & \text{for } \gamma<1 \text{ or } \gamma>1 \end{cases}$                | $M'_1 = \frac{2\beta[(\gamma-1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1-k_1}$                                                                                                                                                       | $\phi'_1 = \frac{\beta}{2(1-k_1)}$       |
| 2.1   | $k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$<br>$D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$ | $M'_{21} = \frac{(2\beta\gamma + C_{21})k_{21}^3 - 3C_{21}k_{21}^2 + 3C_{21}k_{21} - C_{21}}{1-k_{21}}$<br>$C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$                                 | $\phi'_{21} = \frac{\beta}{2(1-k_{21})}$ |
| 2.2   | $k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$<br>$D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$                                          | $M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$<br>$C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$                                                   | $\phi'_{22} = \frac{\beta}{2(1-k_{22})}$ |
| 3.1   | $k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$<br>$D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$        | $M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$<br>$C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$ | $\phi'_{31} = \frac{\beta}{2(1-k_{31})}$ |
| 3.2   | $k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$<br>$D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$          | $M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$<br>$C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$                    | $\phi'_{32} = \frac{\beta}{2(1-k_{32})}$ |

$$M = M' M_{cr}$$

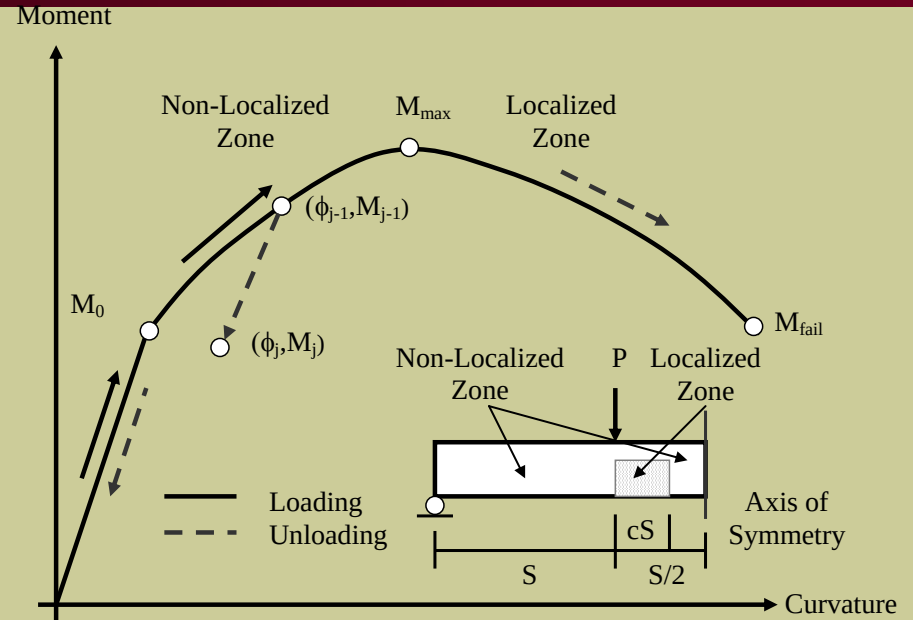
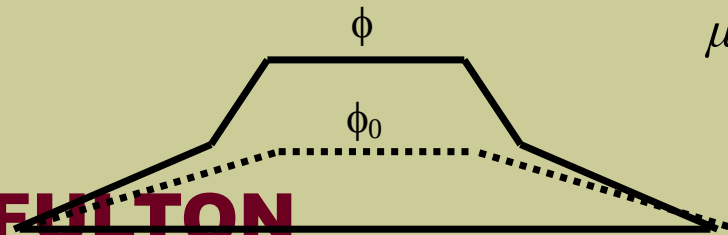
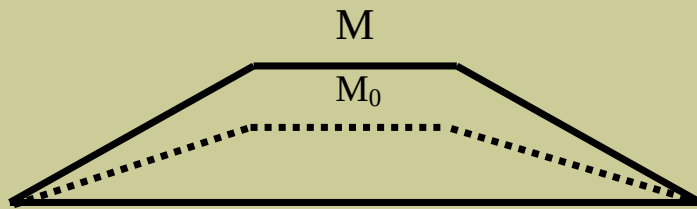
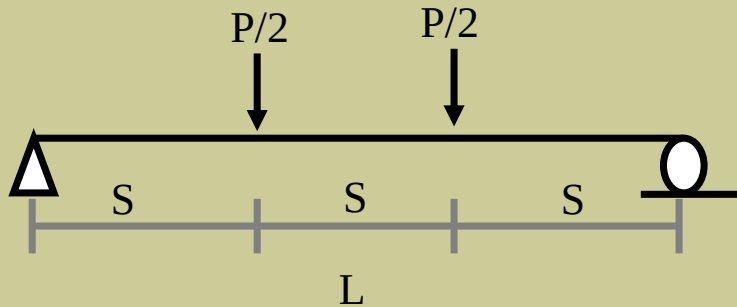
$$M_{cr} = \frac{1}{6}bd^2E\varepsilon_{cr}$$

$$\phi = \phi' \phi_{cr}$$

$$\phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

# Load Deflection Response Applicable to 3PB and 4PB Tests

- Moment area method
- Crack localization rules



$$\delta_{cr} = \frac{23}{216} L^2 \phi_{cr} \text{ in elastic region}$$

$$\mu > \mu_{crit}: \delta_u = \frac{L^2}{24M_u^2} \left[ (2M_u^2 - M_u M_{cr} - M_{cr}^2) \phi_u + (M_u^2 + M_u M_{cr}) \phi_{cr} \right]$$

$$\mu < \mu_{crit} \quad \delta_u = \frac{\phi_u L_p}{8} (2L - L_p) + \frac{M_u \phi_{cr} L}{12M_{cr}} (L - 2L_p)$$



# Back Calculation Parameters

## Model Parameters

### Beam Size

Test Method **4** Point Bending

b = **50**

d = **25**

L = **300**

### Material Model

E = **25000**

$\epsilon_{cr}$  = **1E-04**

$\alpha$  = **40**

### Tension

$\gamma$  = **0.95**

$\eta$  = **0.019**

$\mu$  = **1.73**

$\beta_{tu}$  = **105**

### Compression

$\omega$  = **10.8**

$\lambda_{cu}$  = **40**

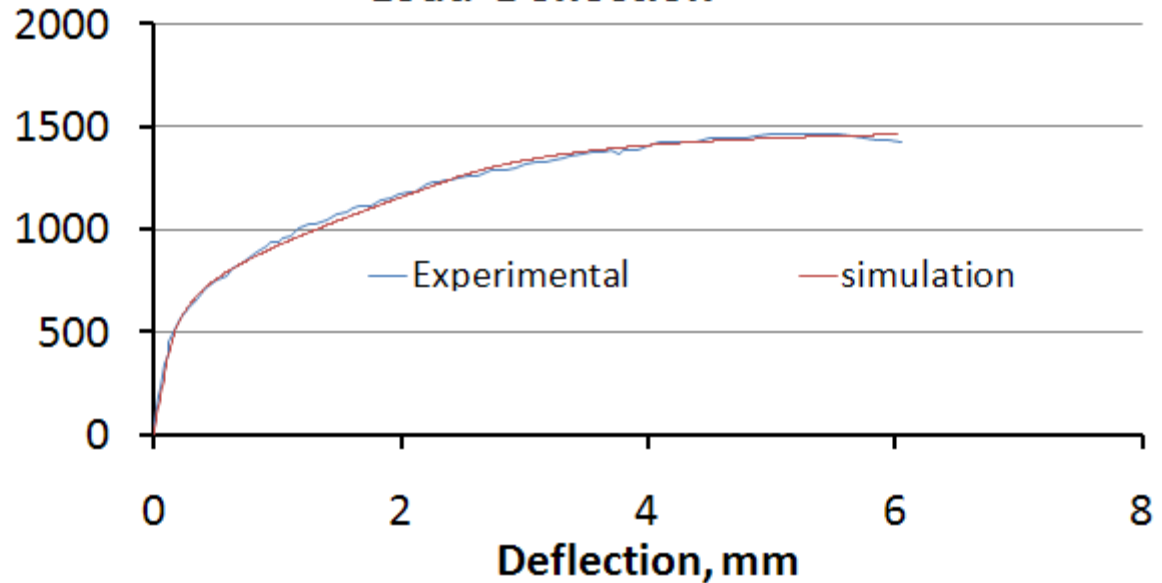
Units are: **in, lbs, psi**

or: **mm, N, Mpa**

Please Use Units Consistently

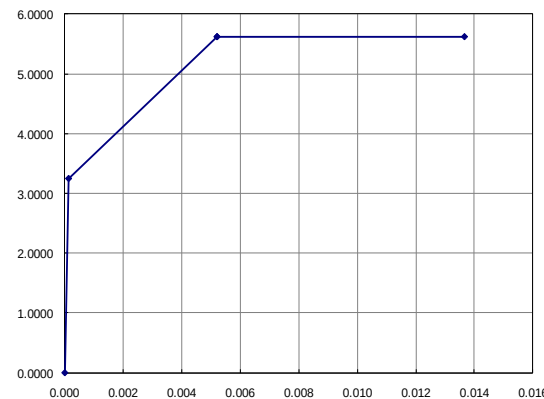
Load, N

## Load Deflection



Tensile Stress

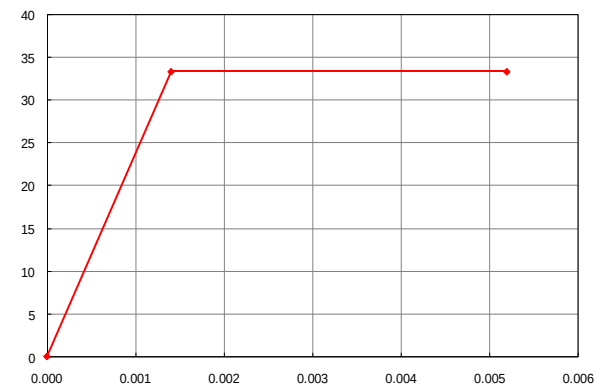
## Tension Model



Tensile Strain

Compressive Stress

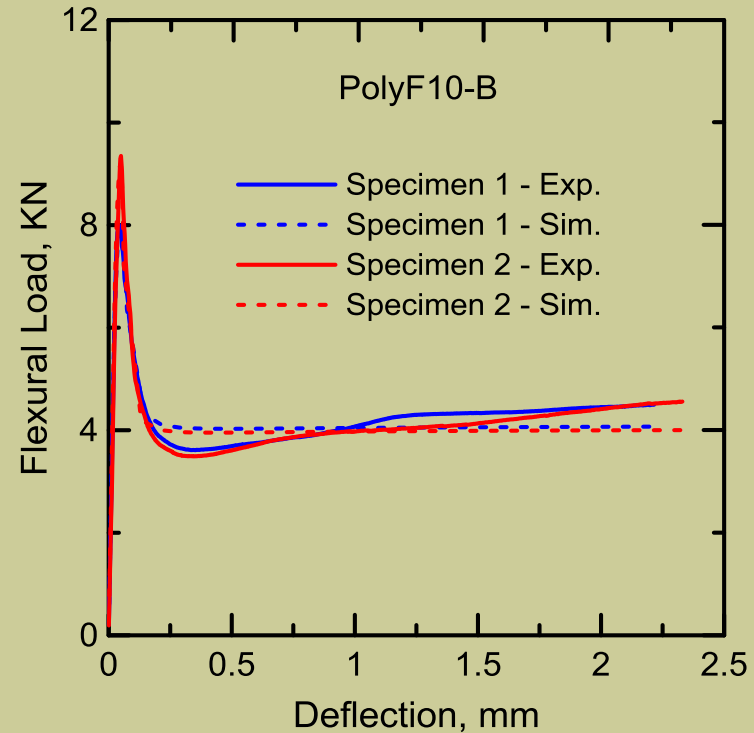
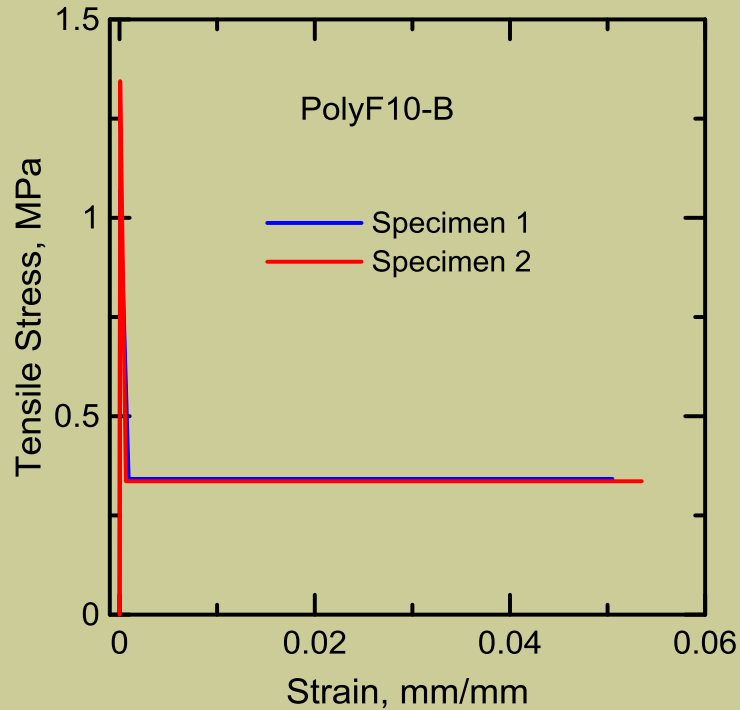
## Compression Model



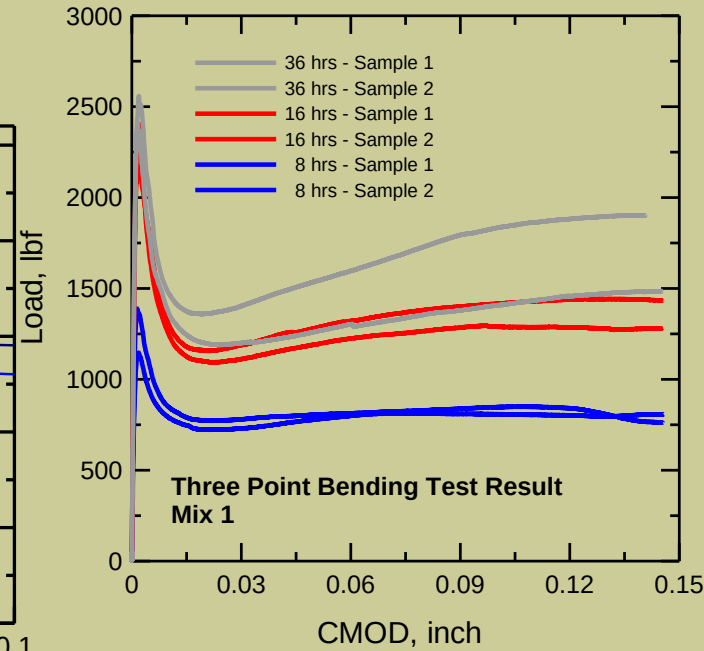
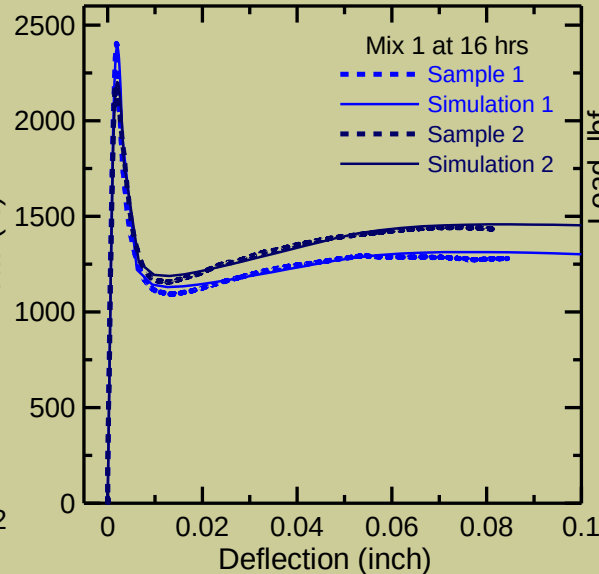
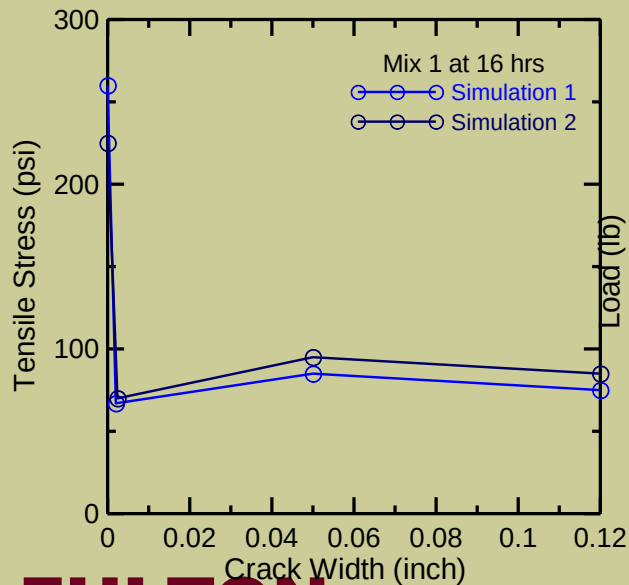
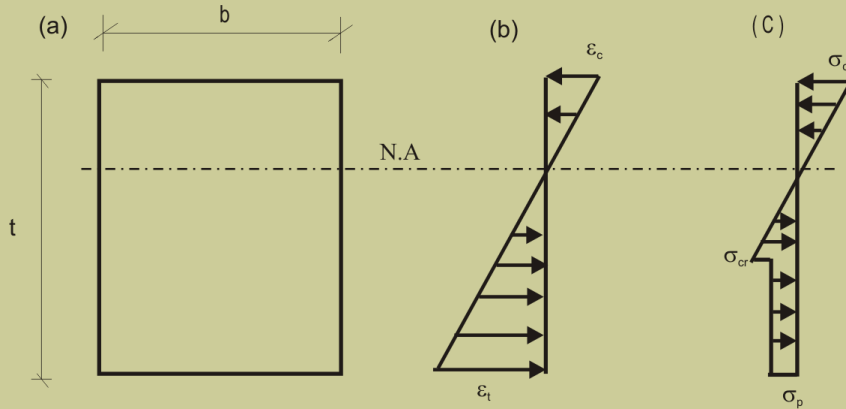
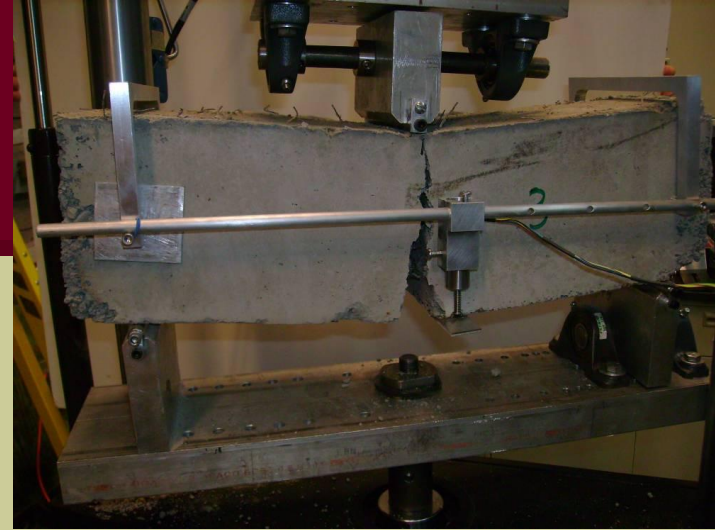
Compressive Strain



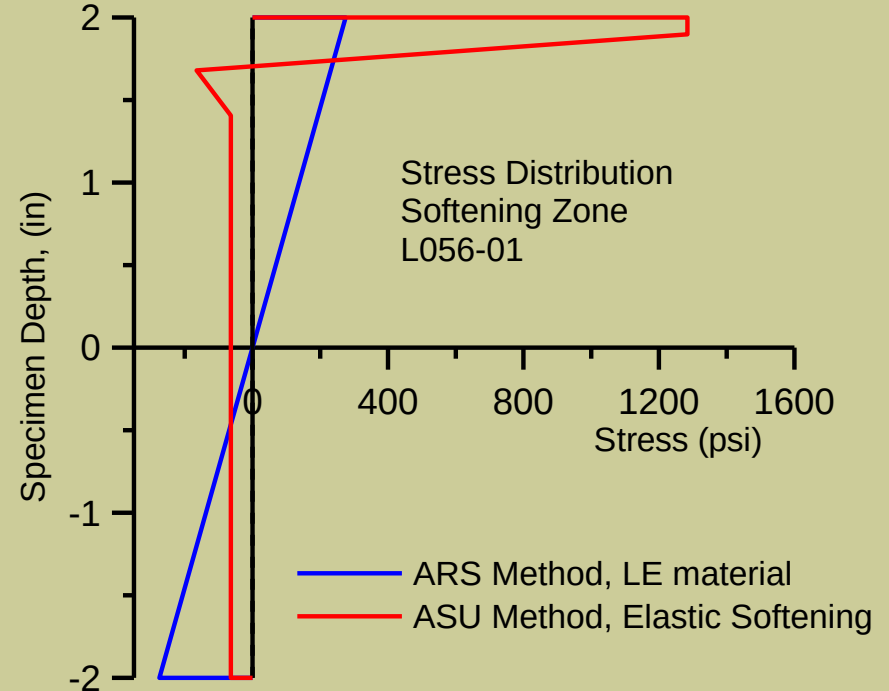
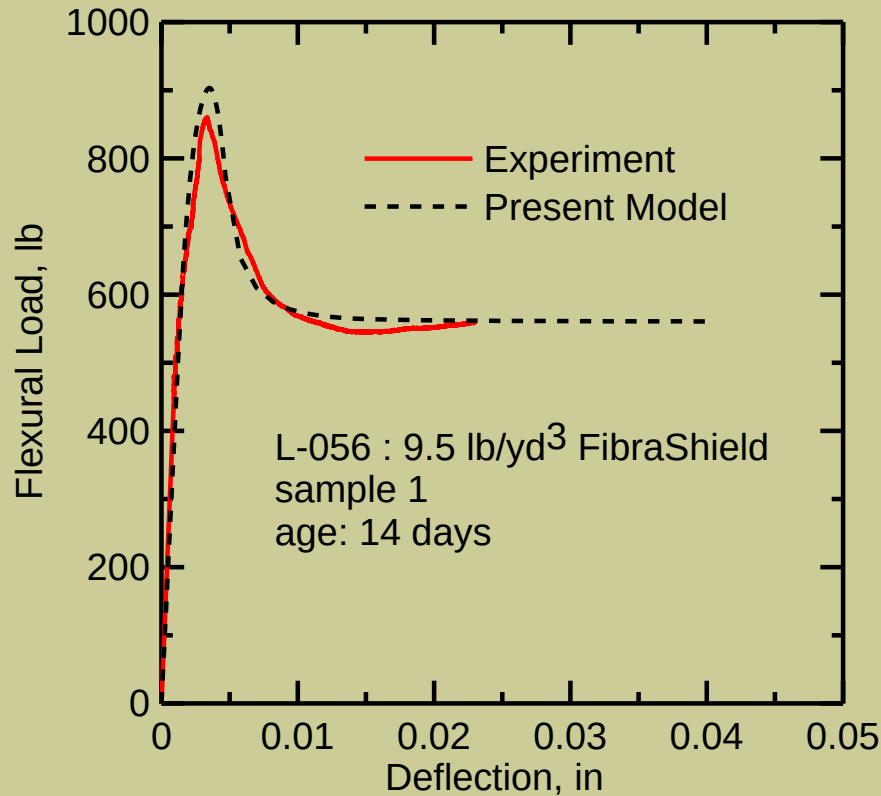
# Back Calculated Material Properties and Tensile Parameters

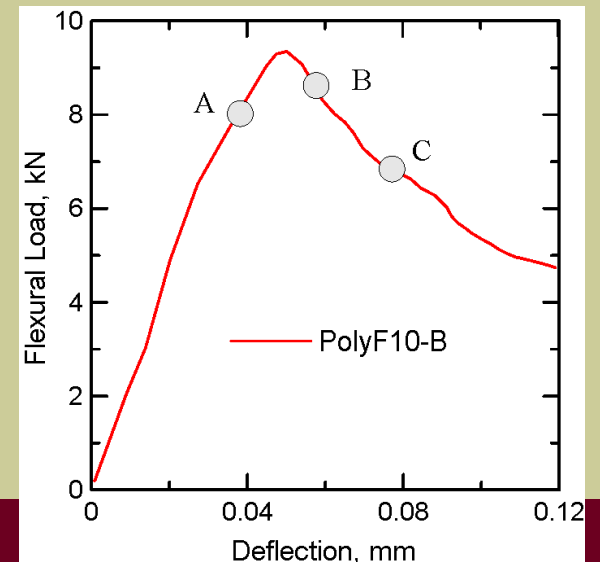
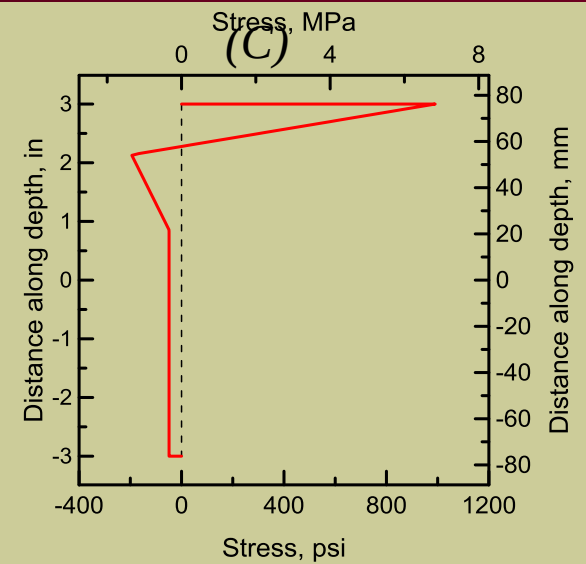
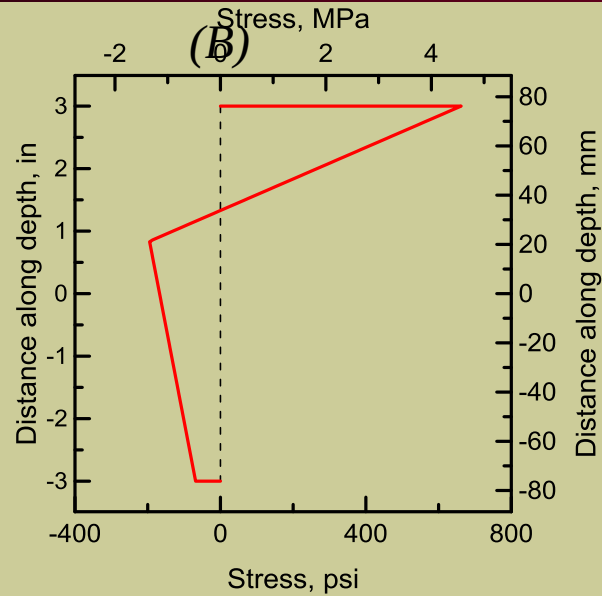
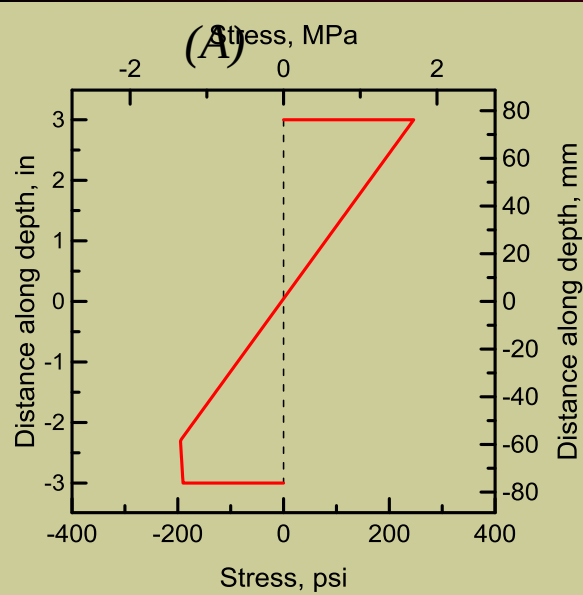


| Mix ID      | $\epsilon_{cr}, \mu\text{str}$ | E, GPa | $\mu$ | $\alpha$ | $\epsilon_{trn}, \mu\text{str}$ | $\epsilon_{tu}, \text{mm/mm}$ |
|-------------|--------------------------------|--------|-------|----------|---------------------------------|-------------------------------|
| PolyF10-A   | 58.5                           | 20.6   | 0.285 | 14       | 793                             | 0.052                         |
| PolyF10-B   | 55                             | 24.1   | 0.37  | 14       | 755                             | 0.050                         |
| PolyF10-C   | 55.5                           | 23.8   | 0.33  | 19       | 1032                            | 0.043                         |
| PolyF10-B&D | 76                             | 17.2   | 0.27  | 9        | 684                             | 0.0495                        |
| PolyF10-C&D | 66                             | 17.2   | 0.39  | 16       | 1056                            | 0.054                         |

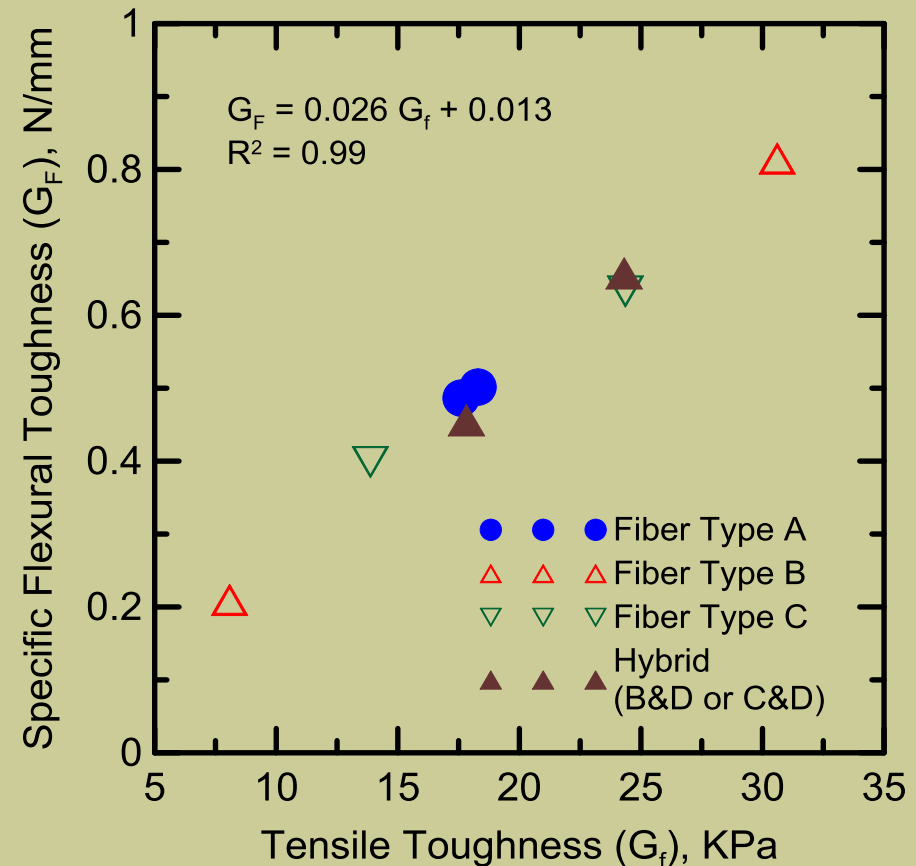
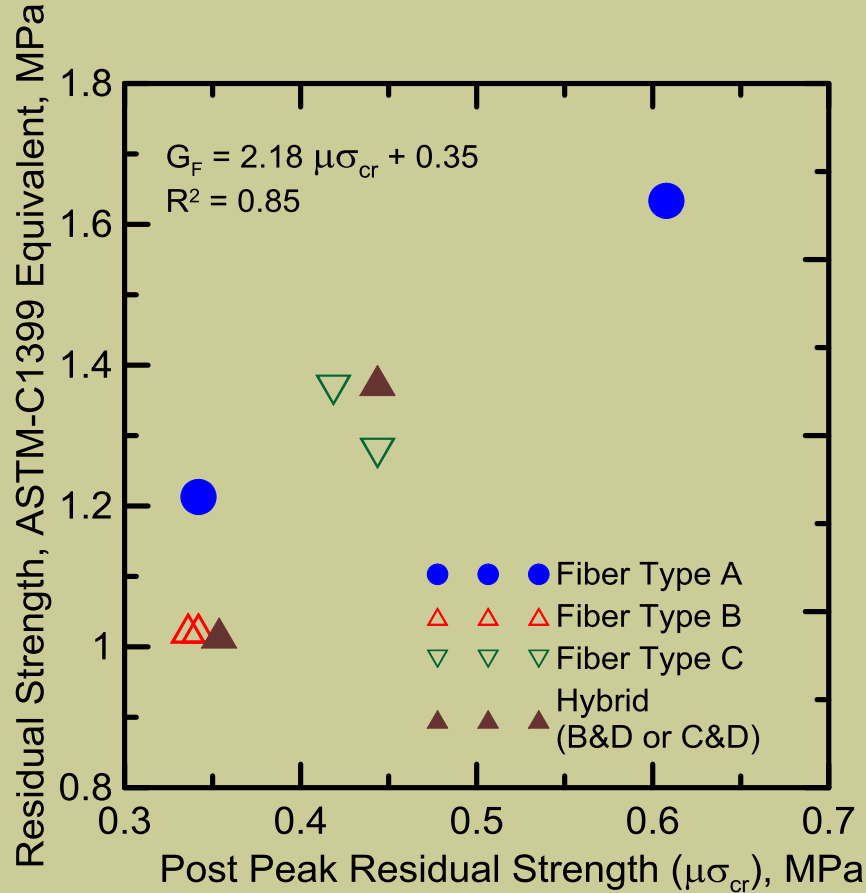


*10-12 lbs/yd<sup>3</sup> of macro fibers*



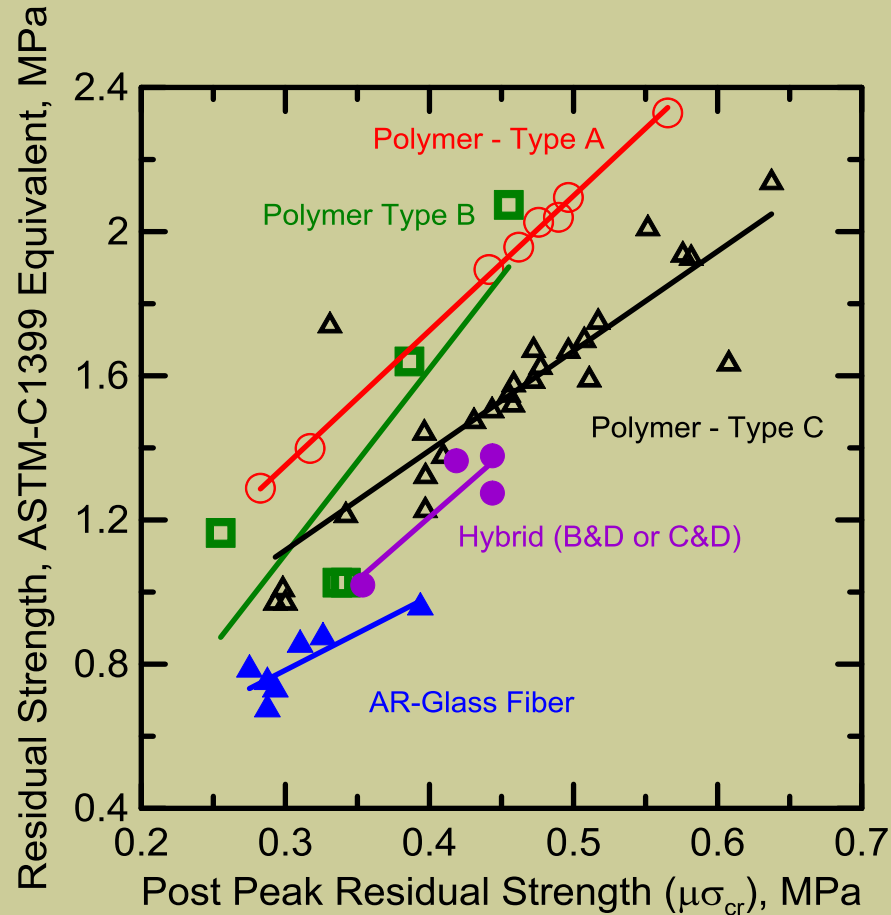


# Post Peak Residual Strength ( $\mu\sigma_{cr}$ ), ARS and Toughness Comparison

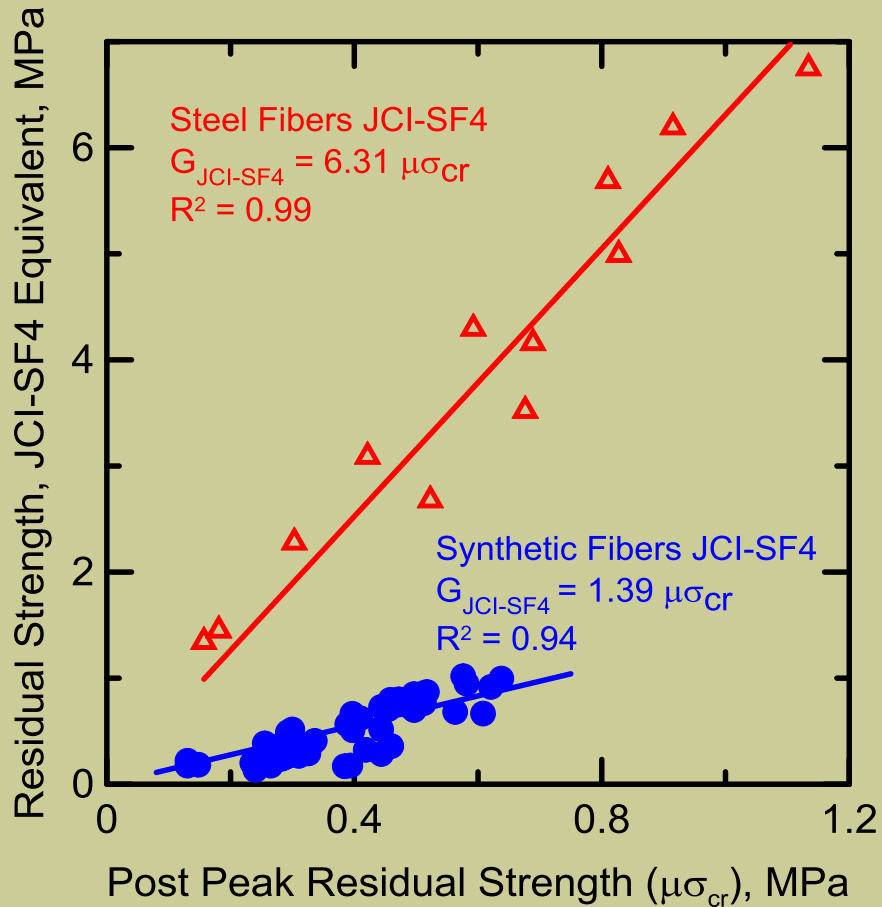


**FULTON** ARS method overestimates the residual tensile strength by 2.2 times  
school of engineering

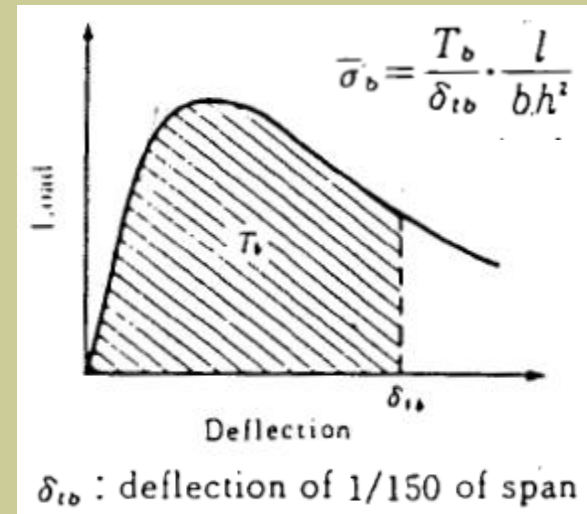
# Curve Fitting of ARS versus Post Peak Residual Strength ( $\mu\sigma_{cr}$ )



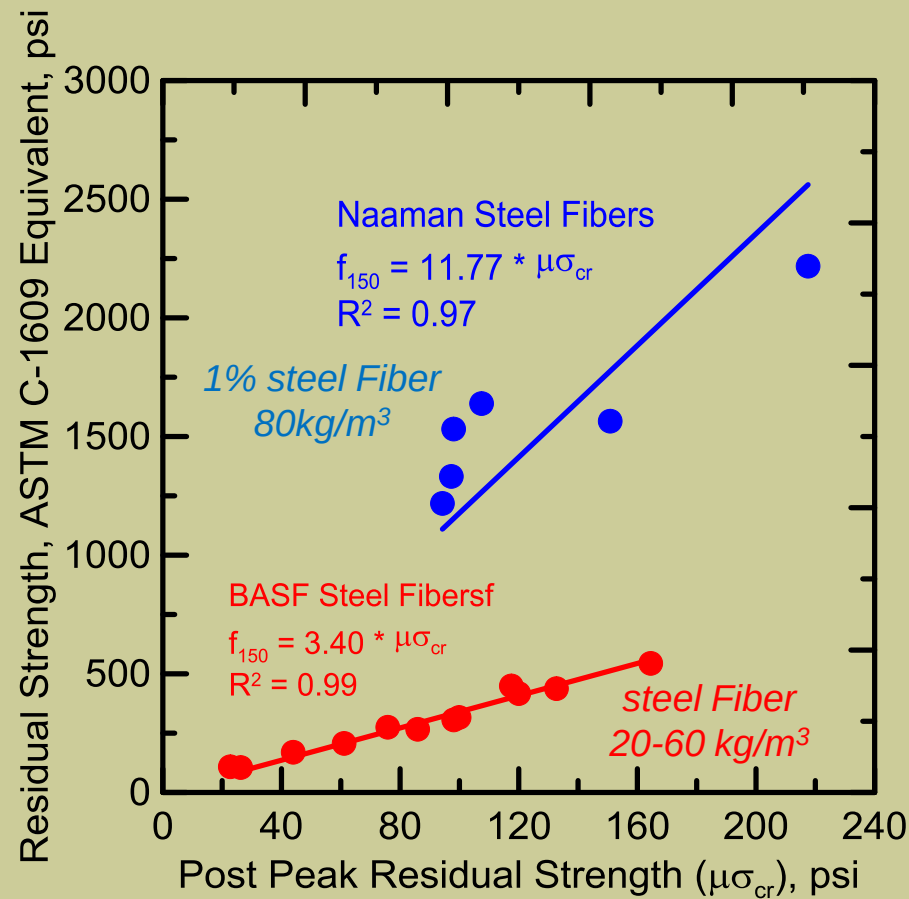
Bakhshi M, Mobasher B. "Sustainable Design of Structural Concrete Materials: a Case Study of Materials Science, Structural Mechanics, and Statistical Process Control", A Report (SR-633) to Arizona Department of Transportation, Tempe, AZ, 2010.



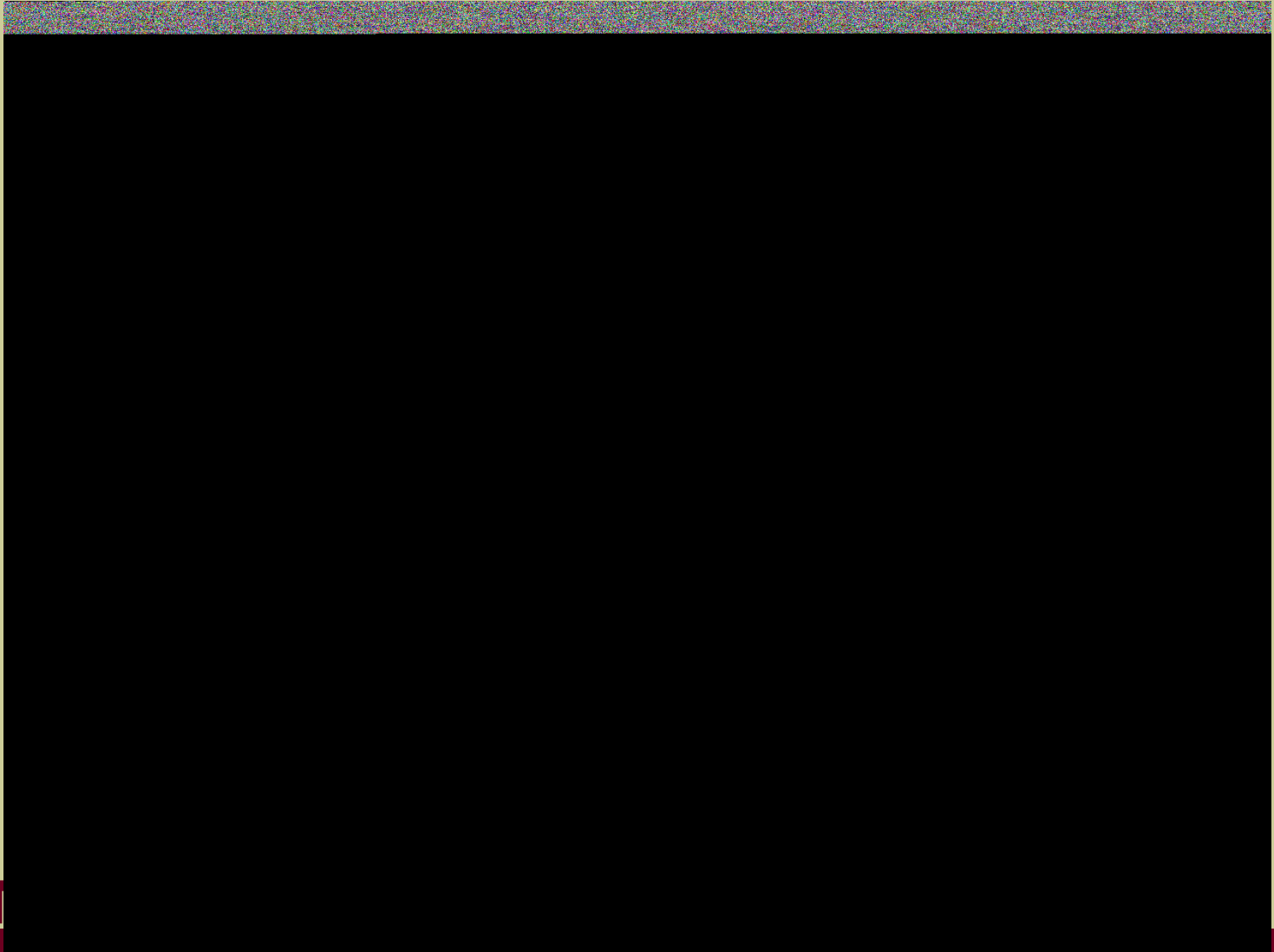
- JCI method overestimates the residual strength of
  - synthetic fibers by 1.4 times
  - steel fibers by 6.3 times



Bakhshi M, Mobasher B. "Sustainable Design of Structural Concrete Materials: a Case Study of Incorporating Materials Science, Structural Mechanics, and Statistical Process Control", A Report (SR-633) to Arizona Department of Transportation, Tempe, AZ, 2010.



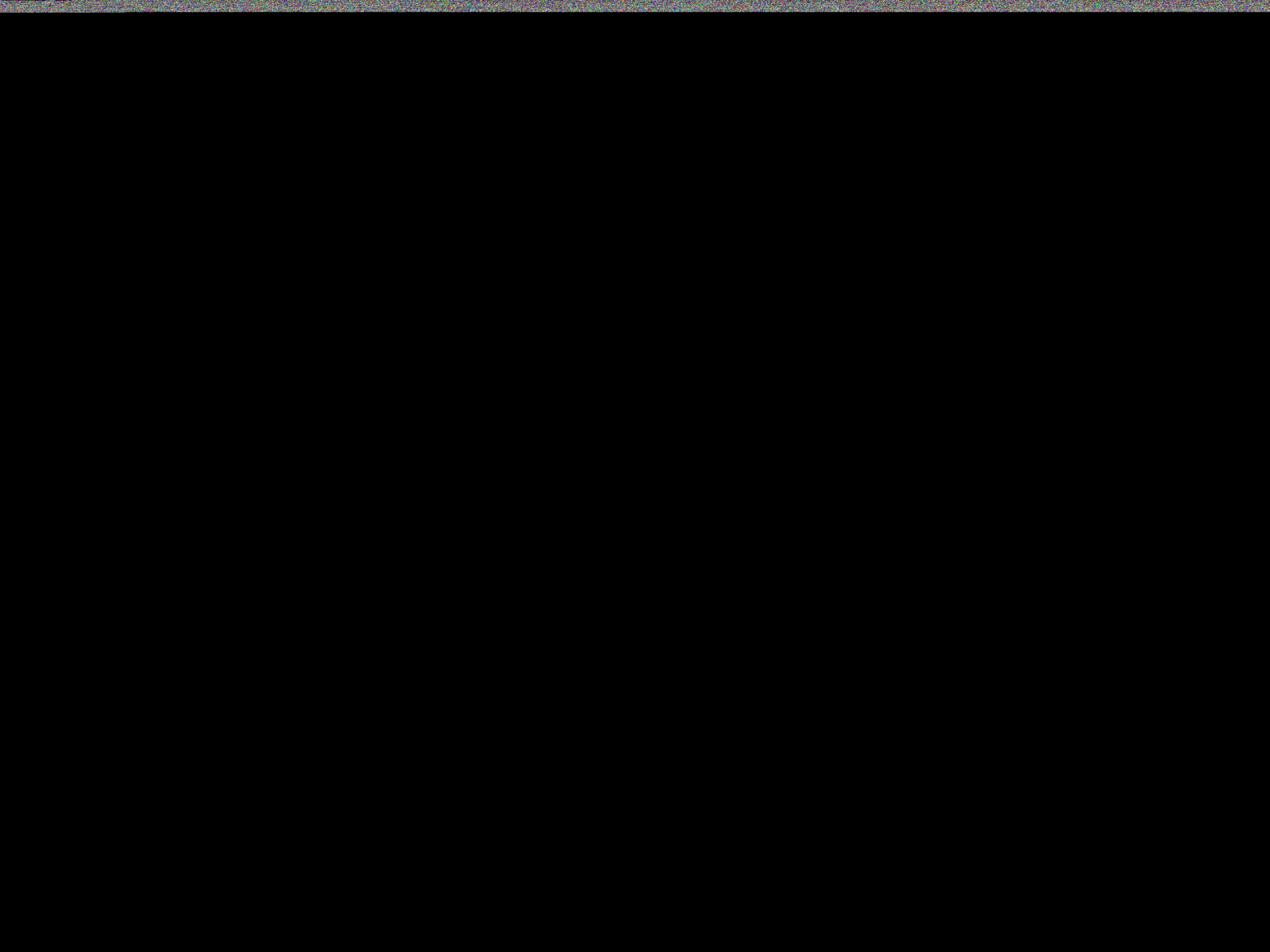
Kim D-J, Naaman AE, El-Tawil S. "Correlation between Tensile and Bending Behavior of FRC Composites with Scale Effect", Proc FraMCoS-7, 7th International Conference on Fracture Mechanics of Concrete and Concrete Structures, May 23-28, 2010, Jeju Island, South Korea





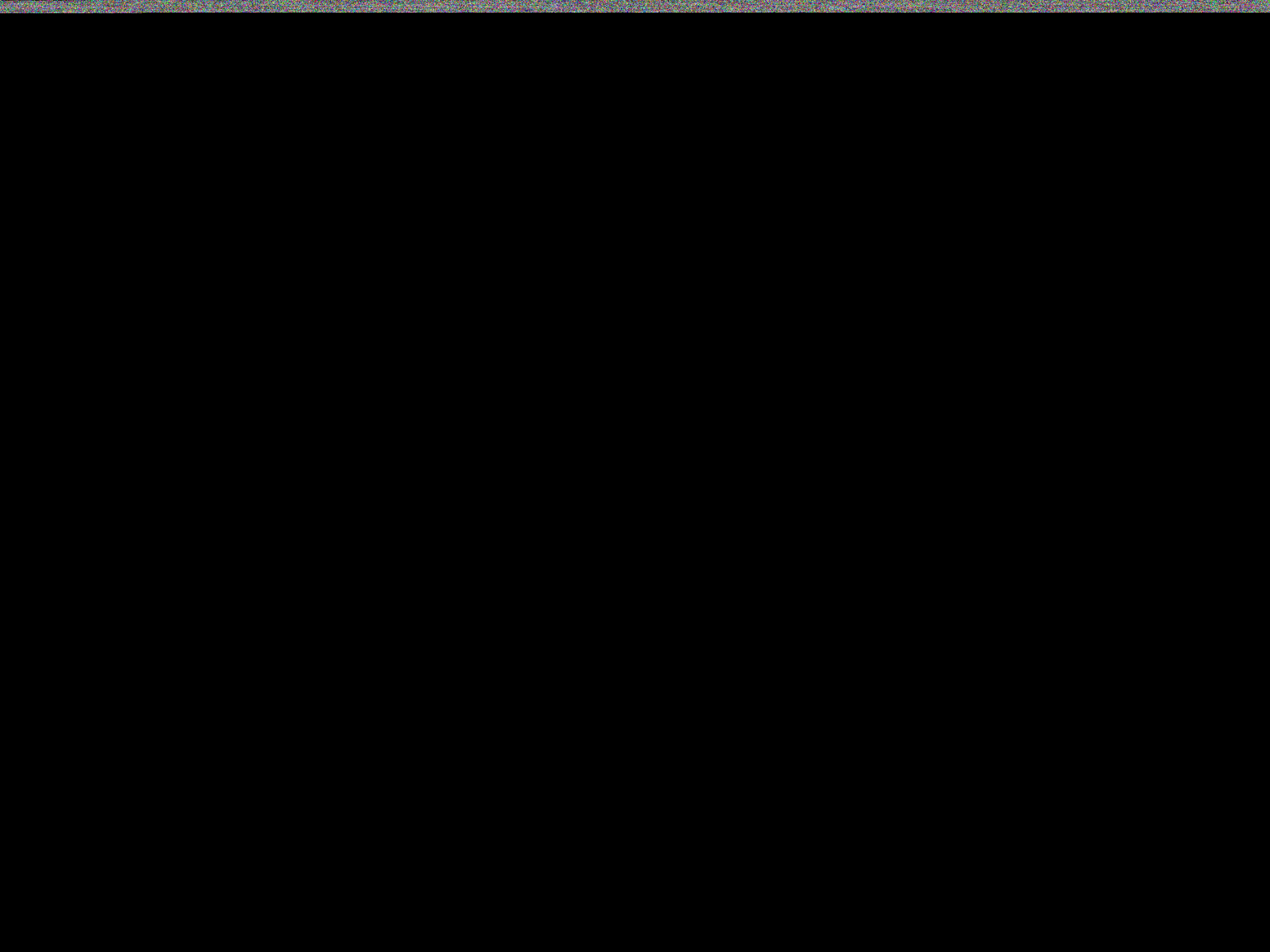
**FULTON**

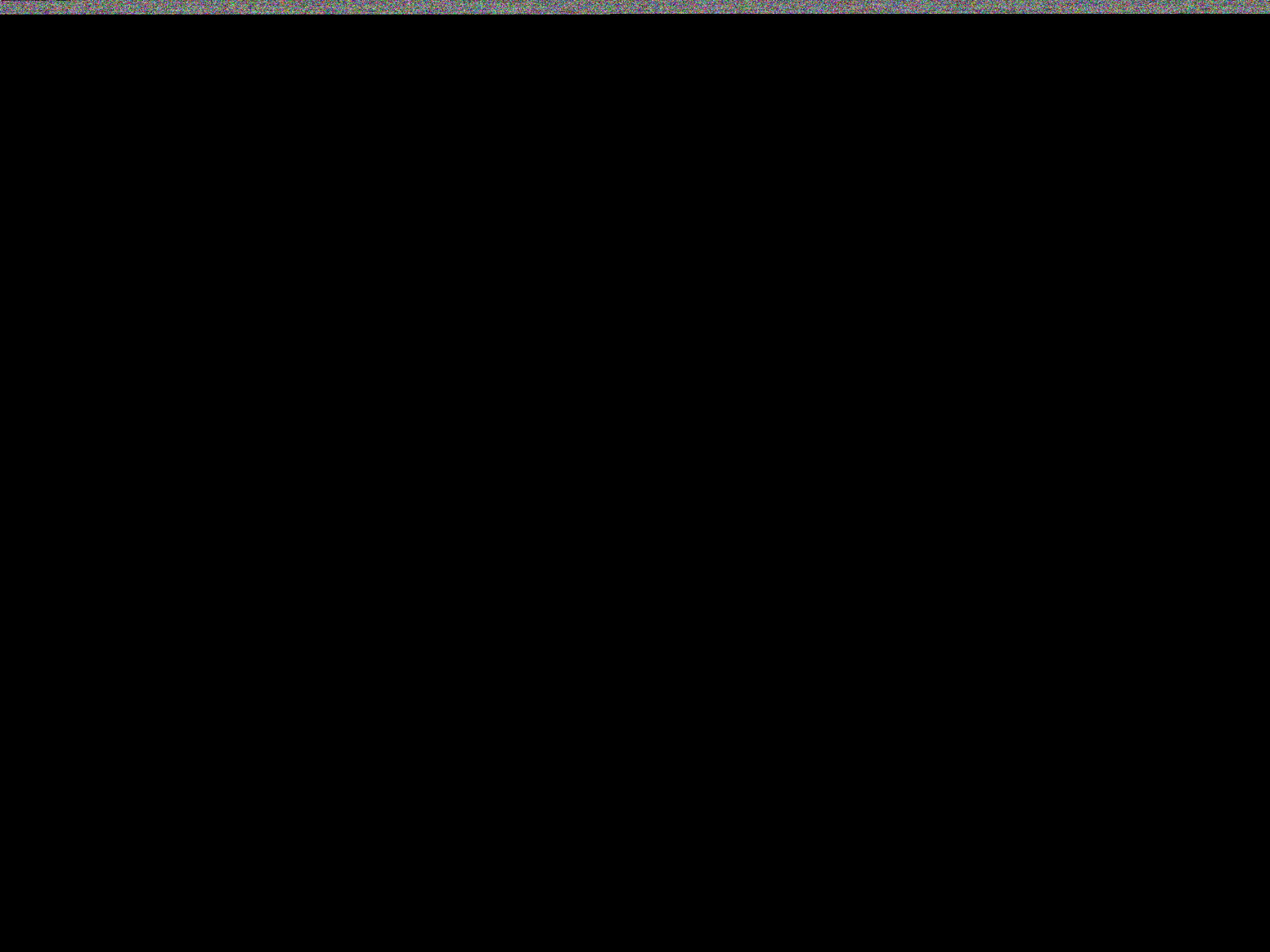
school of engineering



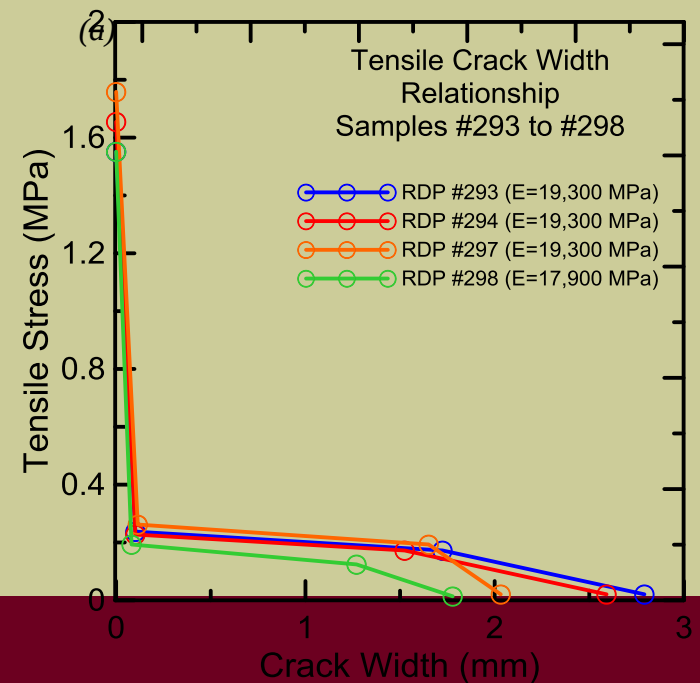
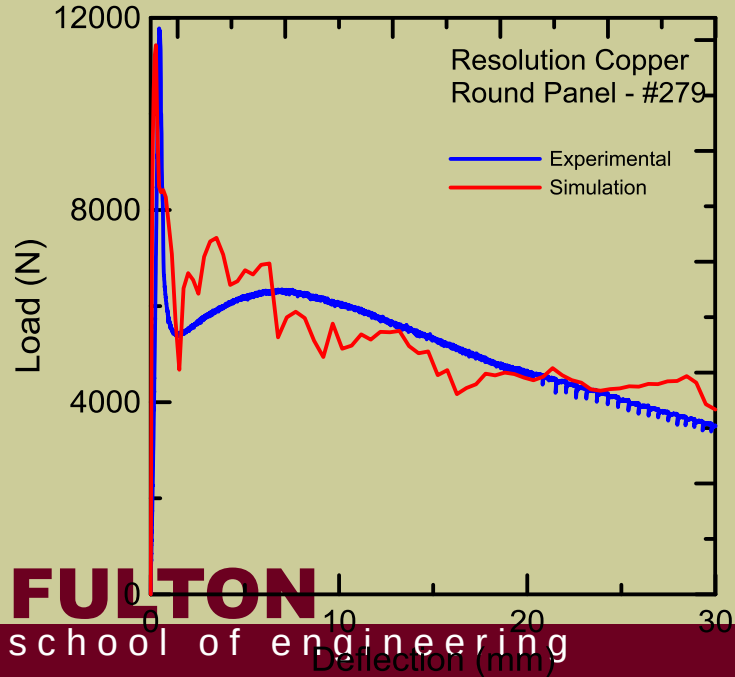
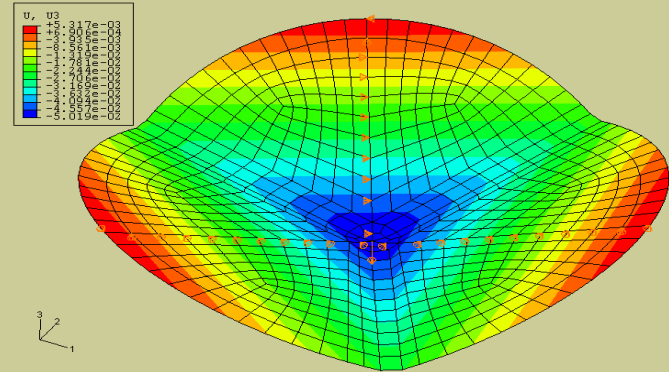






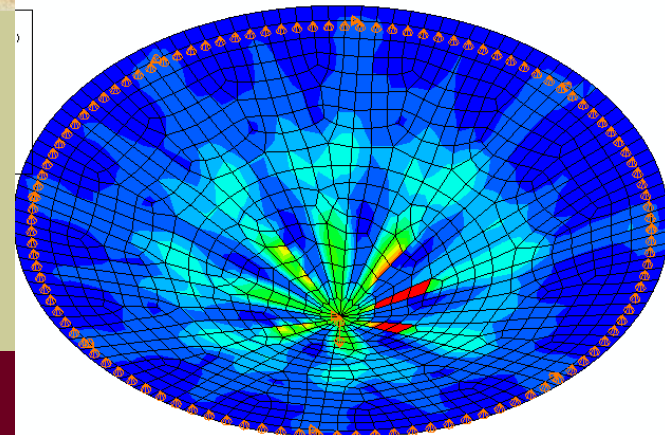
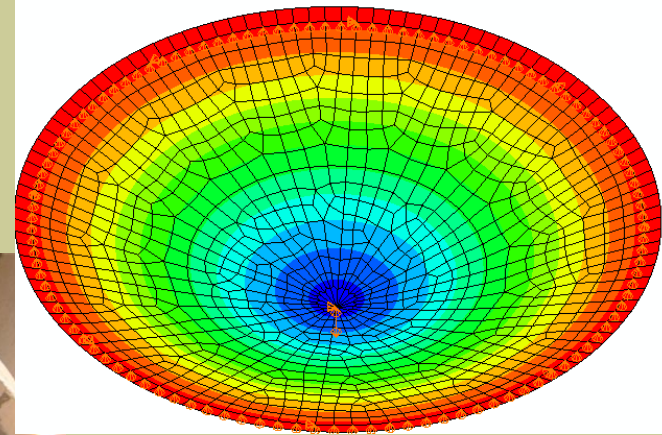
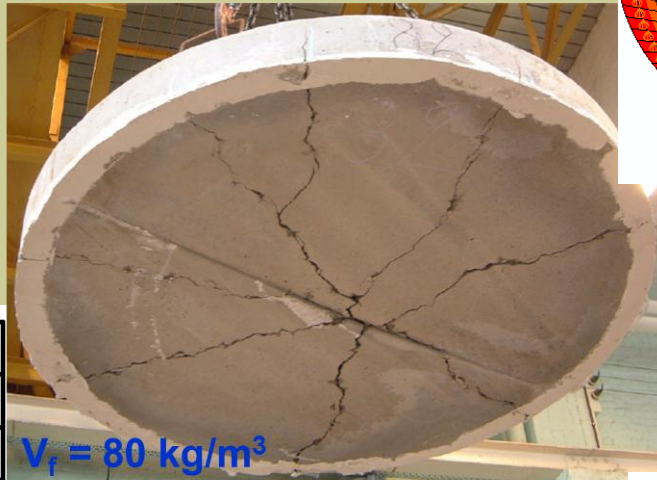
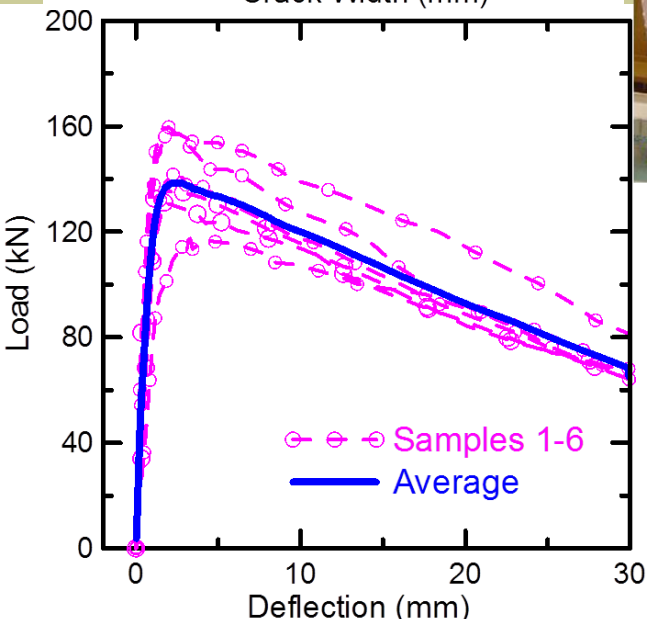
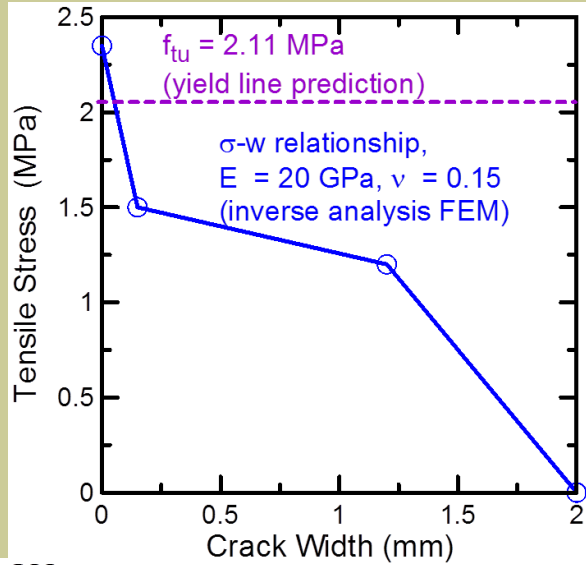


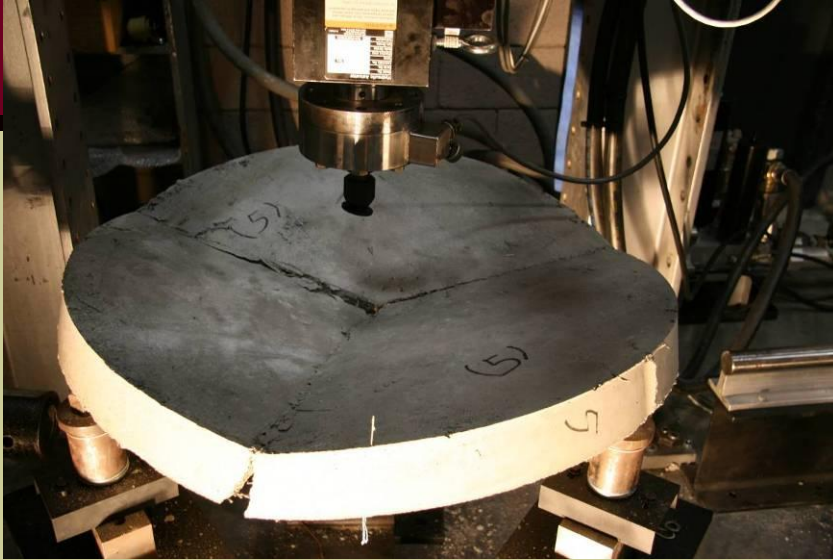
# ASU Panel 3P-support specimen, ASTM C1550





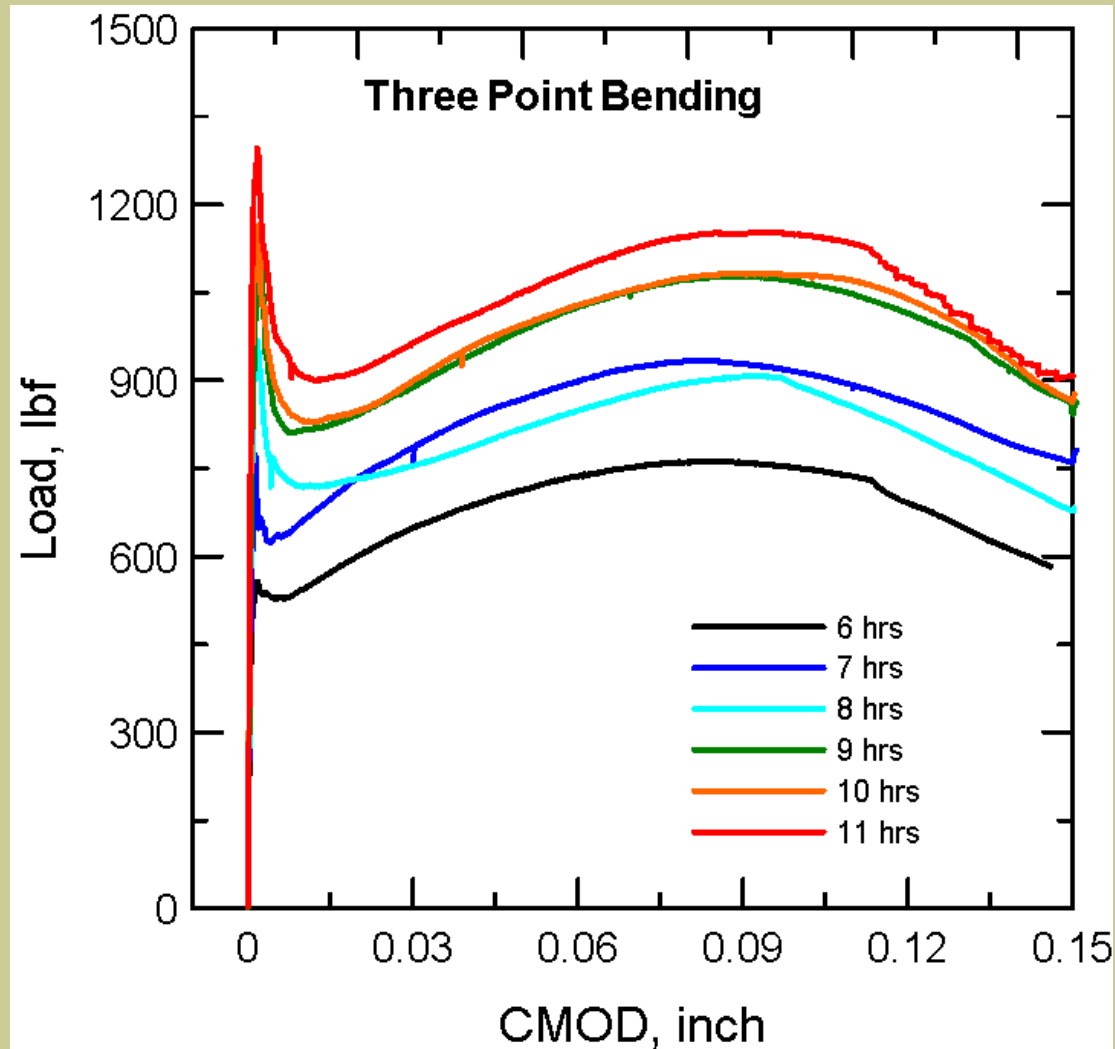
# Round Panel specimens-Continuous support

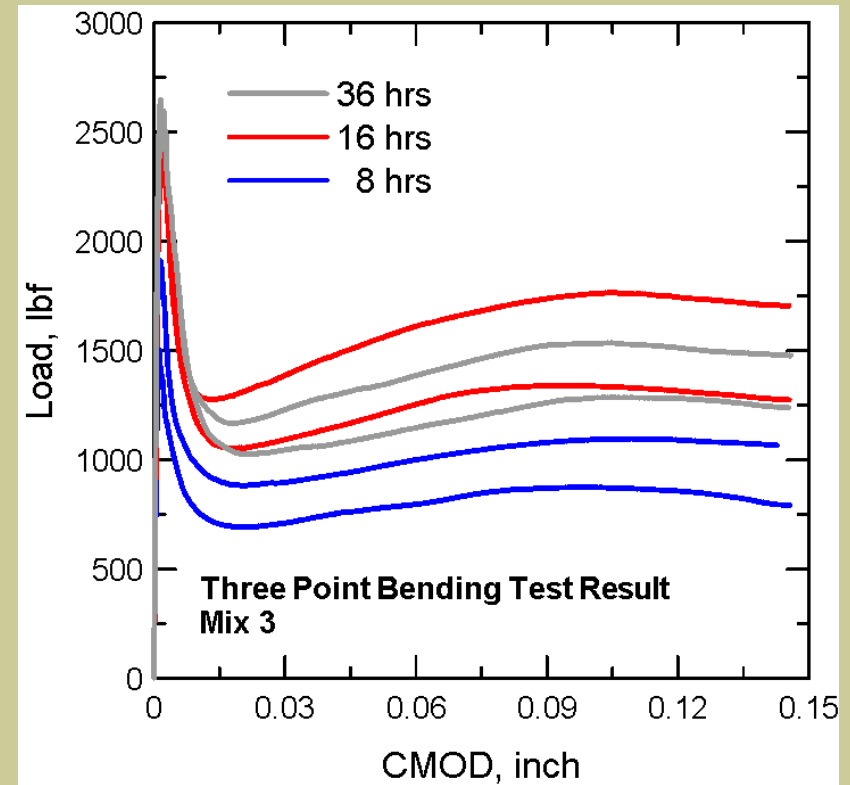
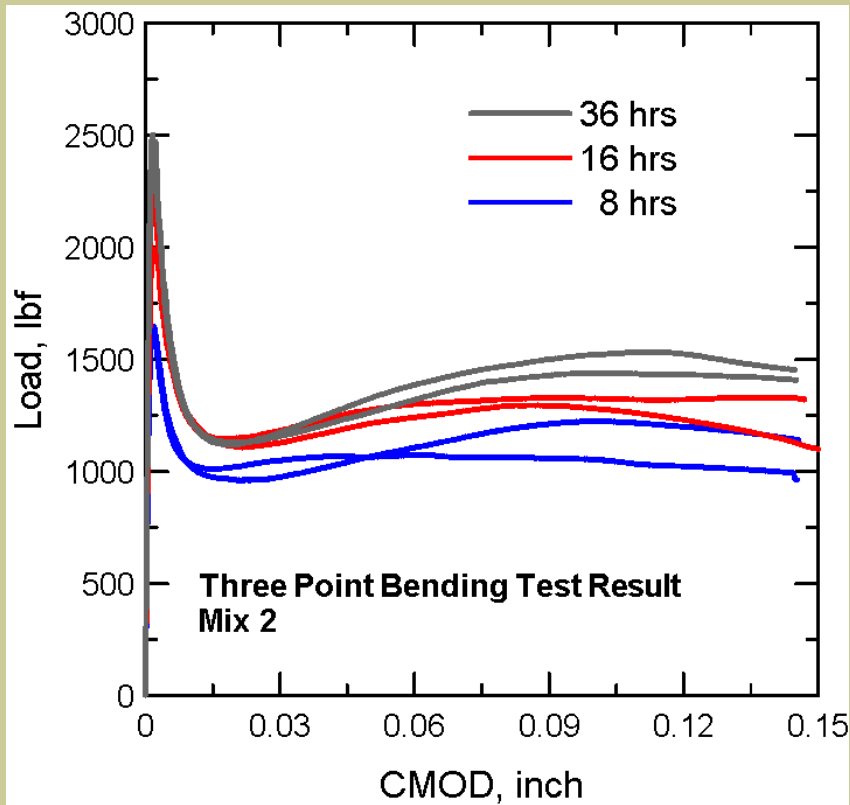




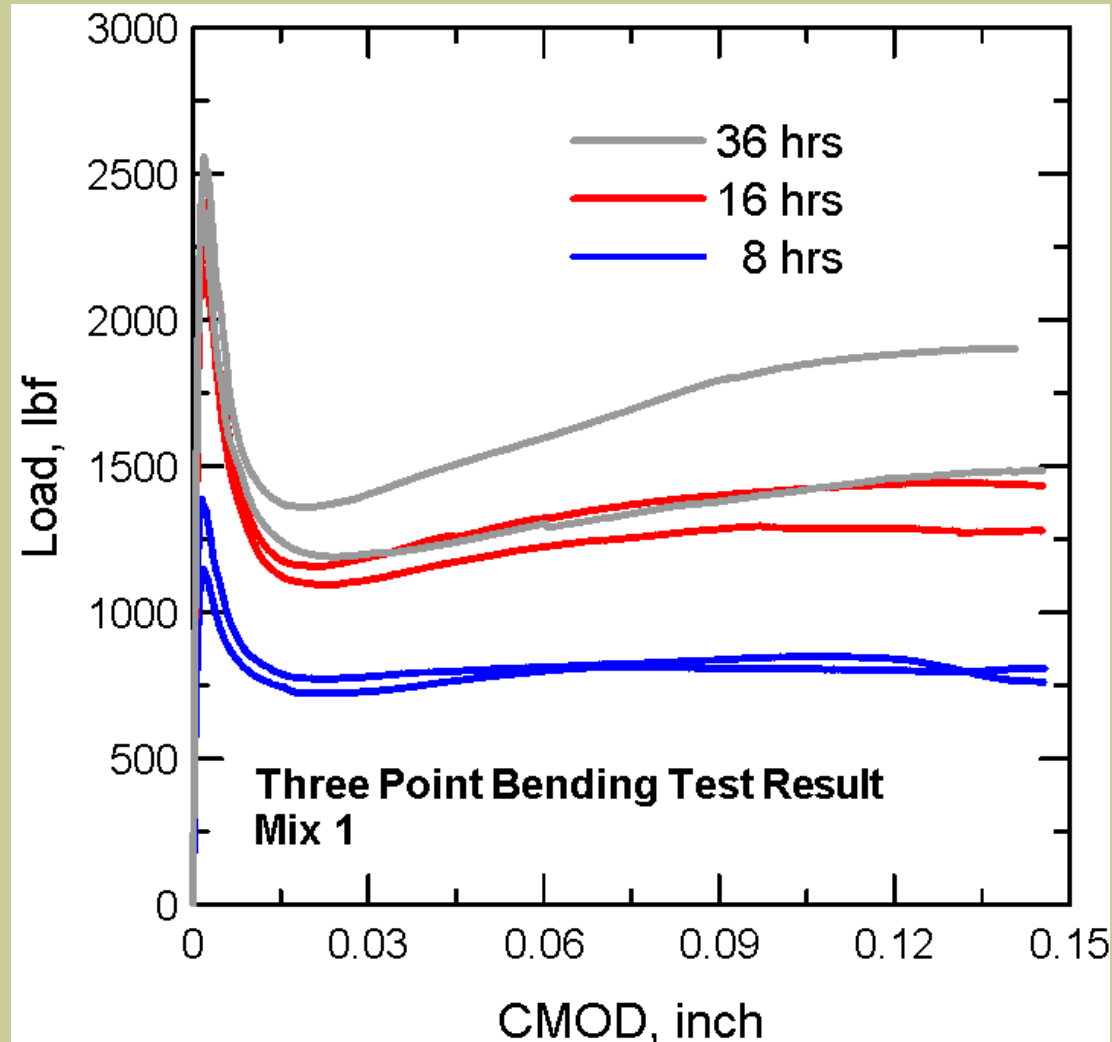
| Item | Sample No. | Id  | Type    | Fiber Type | Fiber Dosage lb/yd <sup>3</sup> | Curing Condition | Age |
|------|------------|-----|---------|------------|---------------------------------|------------------|-----|
| 1    | (dummy)    | I-3 | sprayed | PP         | 5.5                             | moisture, 75°F   | 6+2 |
| 2    | 2          |     | sprayed | PP         | 9.625                           | moisture, 75°F   | 5+2 |
| 3    | 3          | I-5 | sprayed | PP         | 5.5                             | moisture, 75°F   | 6+2 |
| 4    | 4          | I-4 | sprayed | PP         | 5.5                             | moisture, 75°F   | 6+2 |
| 5    | 5          |     | sprayed | PP         | 9.625                           | moisture, 75°F   | 5+2 |
| 6    | 6          |     | sprayed | PP         | 9.625                           | moisture, 75°F   | 5+2 |

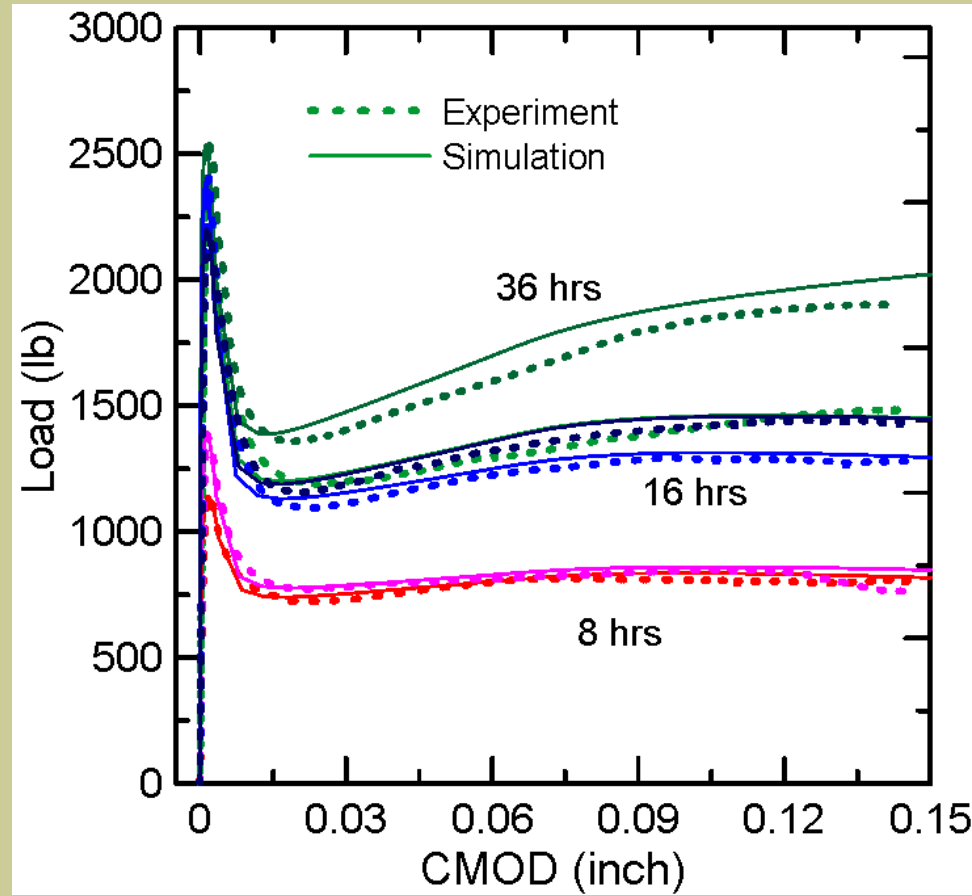
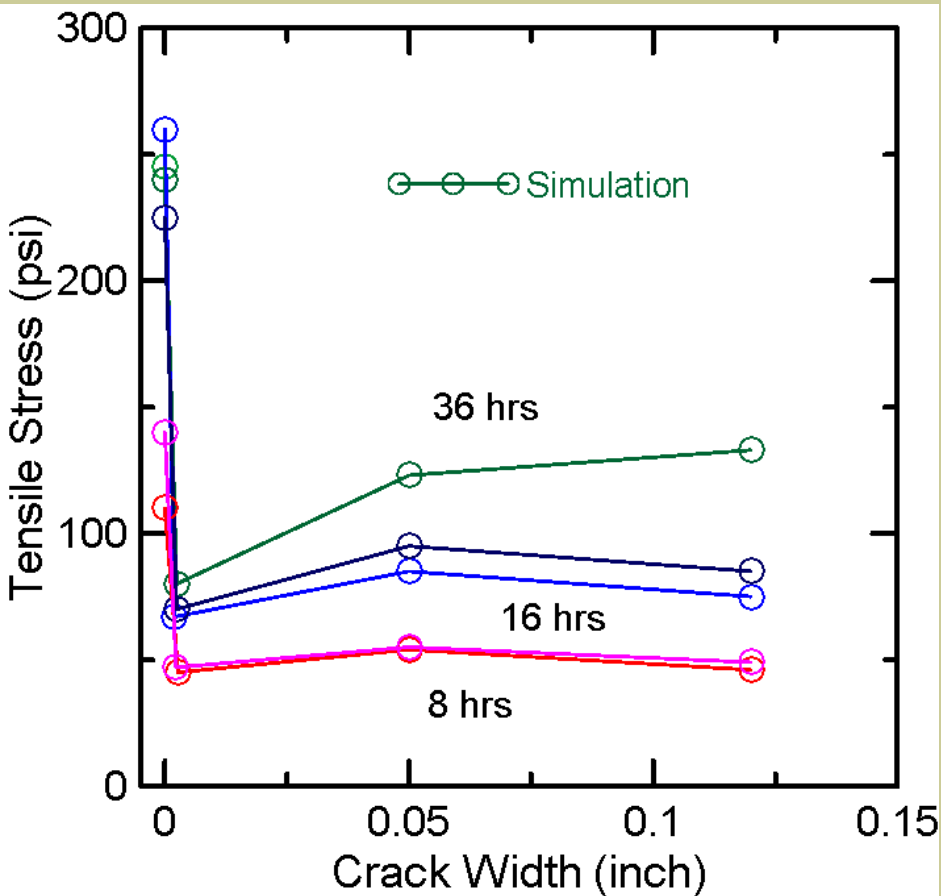
# Gradual strength gain as a function of time



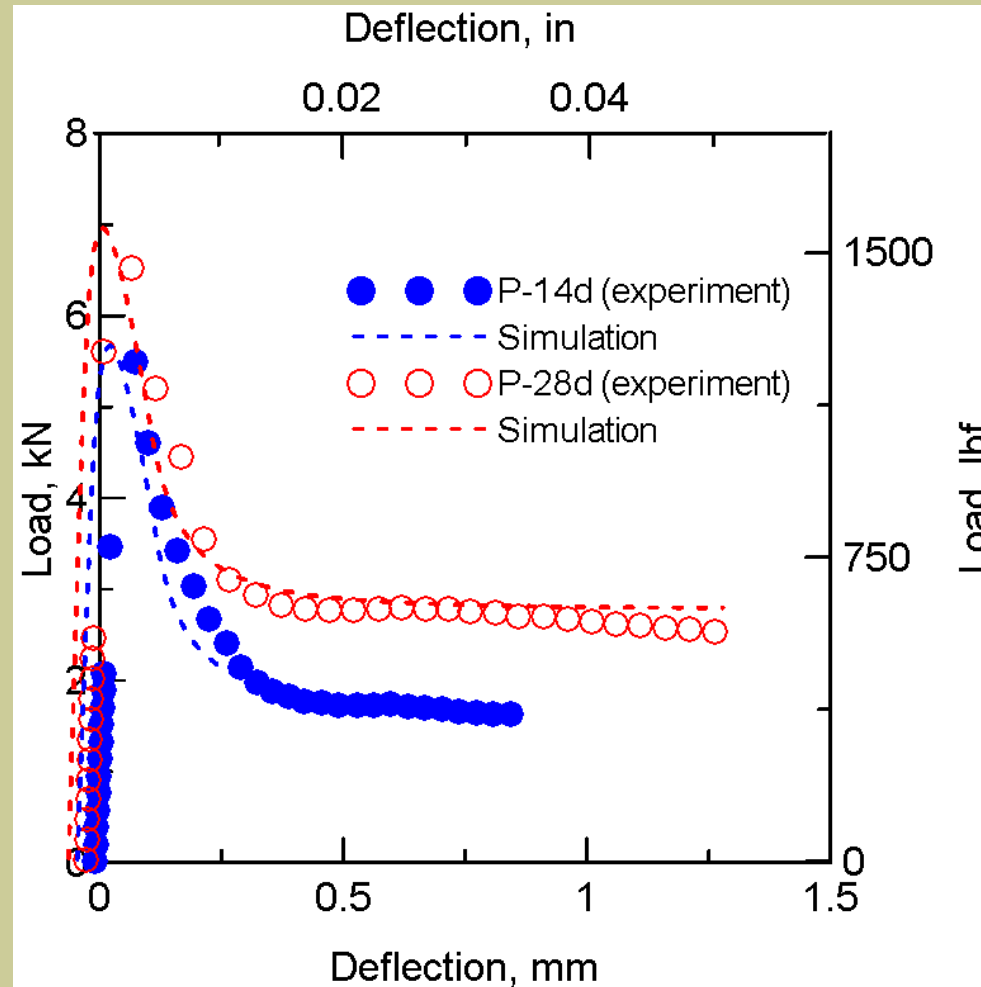


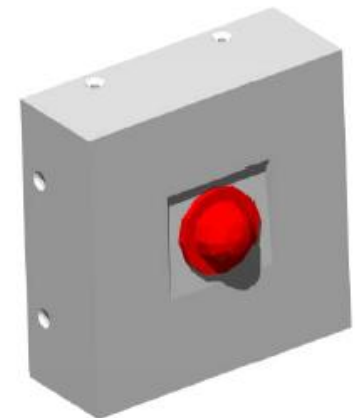
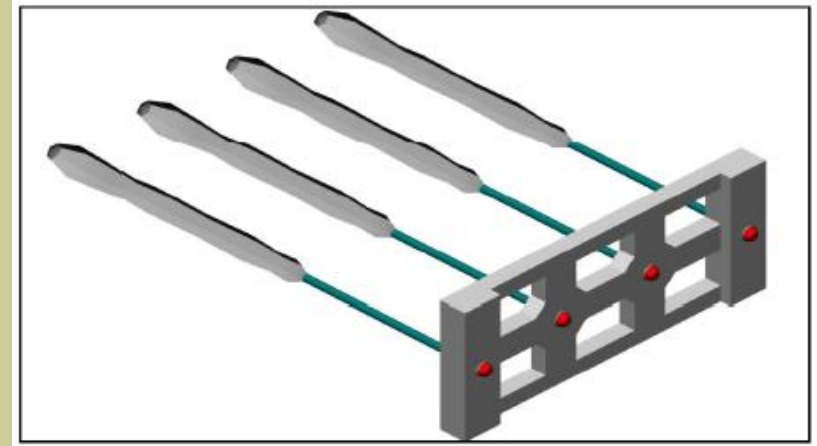
# Effect of Age on the post crack Mechanical response

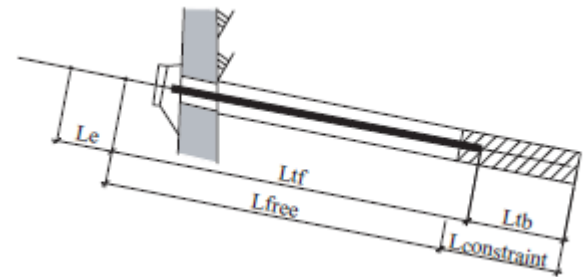




# Effect of curing up to 28 days















# Structural Design with FRC Materials Requires proper guidelines for testing, analysis and Design



*Elevated slabs*

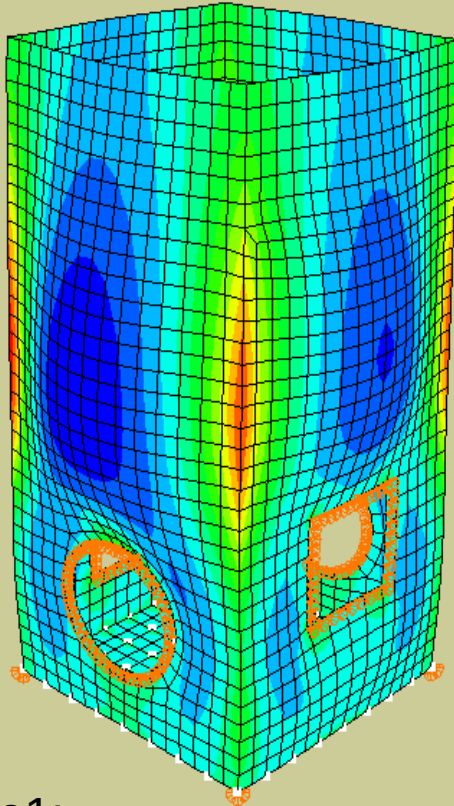
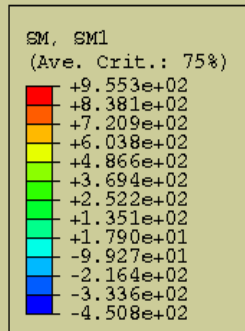
*Precast panels*

*Shotcrete applications*

# Installation of precast water tank

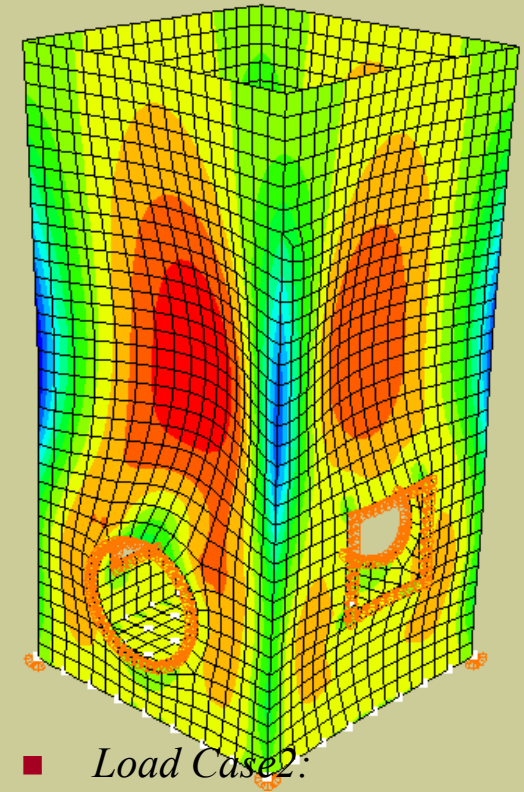
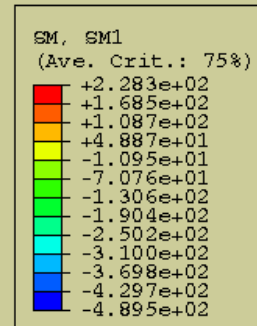
- Panels are assembled on site
- The wall joints are connected using bolts and epoxy
- The base slab is connected to the periphery walls by friction through slots





## ■ Load Case1:

- 1.4 Self weight +
- 1.4 Water pressure



## ■ Load Case2:

- 1.4 Self weight +
- 1.7 Earth pressure +
- 1.7 Uniform pressure due to surcharge
- Moment in short span direction SM1

- Proposed approach for backcalculation of stress strain response is applicable to both Strain hardening and Strain softening Fiber reinforced concrete (FRC)
- Major physical parameters define tensile response of fiber reinforced cementitious materials
- back calculation procedures allow one to obtain tensile response from three-point and four-point bending tests
- The material parameters can be used for design, or finite element analysis of structural members
- Post peak residual strength of FRC is overestimated by available experimental methods such as ARS, JCI and RILEM.



**Stay Safe, be creative**





- Mix 1,2,3,4 are CSA, CSB, HAS, HSB respectively.
- C = control mix, H = high performance mix, S = shogun fiber
- A = fiber dosage level 10 lb/yd<sup>3</sup>, B = fiber dosage level 12 or 15 lb/yd<sup>3</sup>
- Beam is 6" x 6" x 18" (clear span) with a one inch notch at the center.
- Assume localized plastic length  $L_p = 0.5h$ , where  $h$  is reduced depth 5"
- Assume compressive modulus is always elastic. Young's modulus in compression and tension are equal.
- Use load-deflection response instead of load-CMOD to back calculate tensile stress crack width relationship. Using load-CMOD in back calculation gives low  $E$  in the range of 500,000 to 900,000 psi while the load-deflection gives  $E$  in the range of  $1.5 \times 10^6$  to  $4 \times 10^6$  psi.
- The simulations of load-CMOD are also plotted with the experimental load-CMOD.
- Tensile stress crack width evaluated up to 0.15 inch due to the experimental load deflection response is terminated around 0.1 inch. The corresponding fracture energy is therefore evaluated up to 0.15 inch.

|             | Mix 1 | Mix 2 | Mix 3 | Mix 4 | Mix 5 | Mix 6 |
|-------------|-------|-------|-------|-------|-------|-------|
| Shogun      | 10.0  |       |       | 8.5   |       | 10.0  |
| Tufstrand   |       | 10.0  |       |       |       |       |
| Mac 200     |       |       | 10.0  |       | 8.5   |       |
| F100        |       |       |       | 1.5   | 1.5   |       |
|             |       |       |       |       |       |       |
| P900        | 8.0   | 8.0   | 8.0   | 8.0   | 8.0   | 8.0   |
| Delvo       | 1.5   | 1.5   | 1.5   | 1.5   | 1.5   | 1.5   |
| G7101       | 13.0  | 15.0  | 15.0  | 15.0  | 17.0  |       |
| PT1552      |       |       |       |       |       | 13.0  |
| NC534       | 20.0  | 20.0  | 20.0  | 20.0  | 20.0  | 20.0  |
|             |       |       |       |       |       |       |
| Spread      | 19.5  | 17.0  | 21.5  | 21.0  | 22.5  | 21    |
| Unit Wt     | 138.6 | 137.0 | 138.4 | 137.6 | 136.8 | 137.1 |
| Air         | 2.6   | 3.4   | 3.1   |       |       |       |
| Temp        | 73    | 72.0  | 74    | 75    | 70    | 72    |
|             |       |       |       |       |       |       |
| 16 hr Temp  | 95    | 99    | 106   | 103   | 89    | 91    |
| 16 hr Break | 3168  | 3167  | 3320  | 3104  | 2170  | 2873  |
|             |       |       |       |       |       |       |
| 24 hr Temp  | 82    | 82    | 73    | 73    | 73    | 74    |



| Concrete Properties |         |                         |                       |                      |       |                 |                         |                         |                       |
|---------------------|---------|-------------------------|-----------------------|----------------------|-------|-----------------|-------------------------|-------------------------|-----------------------|
| Mix No.             | Code    | Tensile Parameters      |                       |                      |       |                 | Compressive Parameters  |                         |                       |
|                     |         | e <sub>cr</sub> (in/in) | S <sub>cr</sub> (psi) | E <sub>t</sub> (psi) | m     | b <sub>tu</sub> | e <sub>cy</sub> (in/in) | e <sub>cu</sub> (in/in) | S <sub>cu</sub> (psi) |
| Mix 1               | 1-1     | 0.00013                 | 351                   | 2.7E6                | 0.16  | 940*            | 0.001275                | 0.004250                | 3442                  |
|                     | 1-2     | 0.00015                 | 405                   | 2.7E6                | 0.24  | 2200            | 0.001500                | 0.005000                | 4050                  |
|                     | Average | 0.00014                 | 378                   | 2.7E6                | 0.2   | 2200            | 0.001388                | 0.004625                | 3746                  |
| Mix 2               | 2-1     | 0.000105                | 283.5                 | 2.7E6                | 0.2   | 3200            | 0.0011250               | 0.003750                | 3037                  |
|                     | 2-2     | 0.000115                | 310.5                 | 2.7E6                | 0.18  | 2900            | 0.0012750               | 0.004250                | 3442                  |
|                     | Average | 0.00011                 | 297                   | 2.7E6                | 0.19  | 3050            | 0.001200                | 0.004000                | 3239                  |
| Mix 3               | 3-1     | 0.00014                 | 378                   | 2.7E6                | 0.19  | 1500*           | 0.0013500               | 0.004500                | 3645                  |
|                     | 3-2     | 0.000135                | 364.5                 | 2.7E6                | 0.19  | 2450            | 0.0013500               | 0.004500                | 3645                  |
|                     | Average | 0.000138                | 371.25                | 2.7E6                | 0.19  | 2450            | 0.0013500               | 0.004500                | 3645                  |
| Mix 4               | 4-1     | 0.000125                | 337.5                 | 2.7E6                | 0.19  | 2650            | 0.0012000               | 0.004000                | 3240                  |
|                     | 4-2     | 0.000125                | 337.5                 | 2.7E6                | 0.15  | 2650            | 0.0011250               | 0.003750                | 3037                  |
|                     | Average | 0.000125                | 337.5                 | 2.7E6                | 0.17  | 2650            | 0.001163                | 0.003875                | 3138                  |
| Mix 5               | 5-1     | 0.000115                | 310.5                 | 2.7E6                | 0.23  | 2800            | 0.001200                | 0.004000                | 3240                  |
|                     | 5-2     | 0.00011                 | 297                   | 2.7E6                | 0.18  | 3000            | 0.001125                | 0.003750                | 3037                  |
|                     | Average | 0.000113                | 303.75                | 2.7E6                | 0.205 | 2900            | 0.001163                | 0.003875                | 3138                  |

